MACHINE LEARNING / MODULE 3

Clustering and mixture models

K-means, density, hierarchy, and probabilistic membership for energy data

Full day · 09:00–17:00 | 6 teaching parts | 3 laboratories | Python · scikit-learn

Machine learning for energy applications

Today's plan

09:00	Why cluster energy data	Lecture
09:15	Part 1: Similarity and scaling	Lecture
09:45	Part 2: K-means	Lecture
10:30	Break	Break
10:45	Lab 1: K-means on well W5	Lab
11:35	Part 3: Density clustering	Lecture
12:15	Lunch	Lunch
13:15	Lab 2: DBSCAN on logs	Lab
14:00	Part 4: Hierarchy, zonation	Lecture
14:40	Break	Break
14:55	Part 5: Gaussian mixtures	Lecture
15:35	Part 6: Validation	Lecture
16:05	Lab 3: Mixtures on rocks	Lab
16:45	Recap and exit check	Discussion

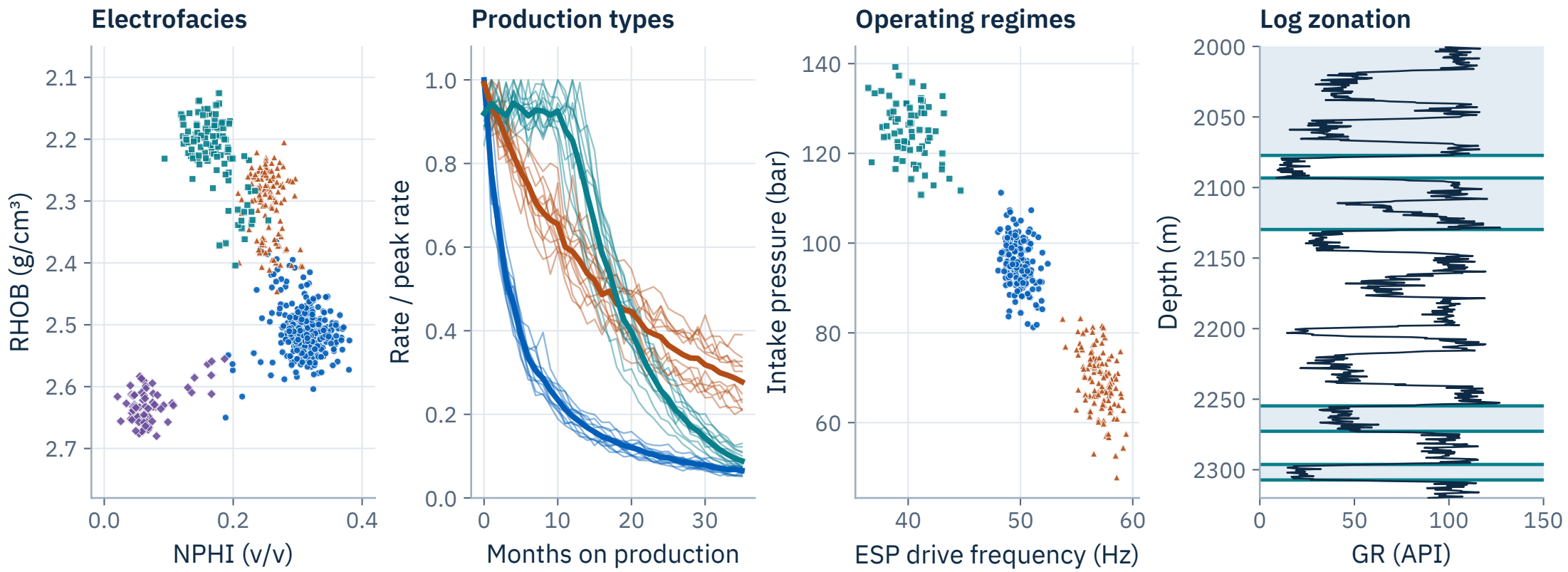
What you will be able to do

By the end of the day you will be able to:

1. Choose features, transforms and scaling that encode the similarity you intend.
2. Run and diagnose K-means: seeding, restarts and the choice of K.
3. Tune DBSCAN or HDBSCAN with a k-distance plot and interpret noise.
4. Build, cut and compare dendrograms, including depth-constrained zonation.
5. Fit Gaussian mixtures and choose K and covariance type with BIC.
6. Validate with stability and external labels; profile clusters in original units.

Clustering finds structure nobody has labelled yet

Four recurring energy tasks where the groups are not known in advance.



Synthetic: well S-1 logs (K-means, K = 4; constrained Ward), decline curves (K-means, K = 3), ESP data (mixture, K = 3).

Every grouping should serve a named decision

Electrofacies

- Rock-type candidates from logs
- Guides core sampling and petrophysical models

Production types

- Wells with similar decline shapes
- Supports type curves and analogue selection

Operating regimes

- Normal, ramped and restart states
- Regime-specific baselines and alarms

Zonation

- Contiguous depth intervals
- Layering for models and completions

◆ Write the decision down before choosing features: it defines what "similar" should mean.

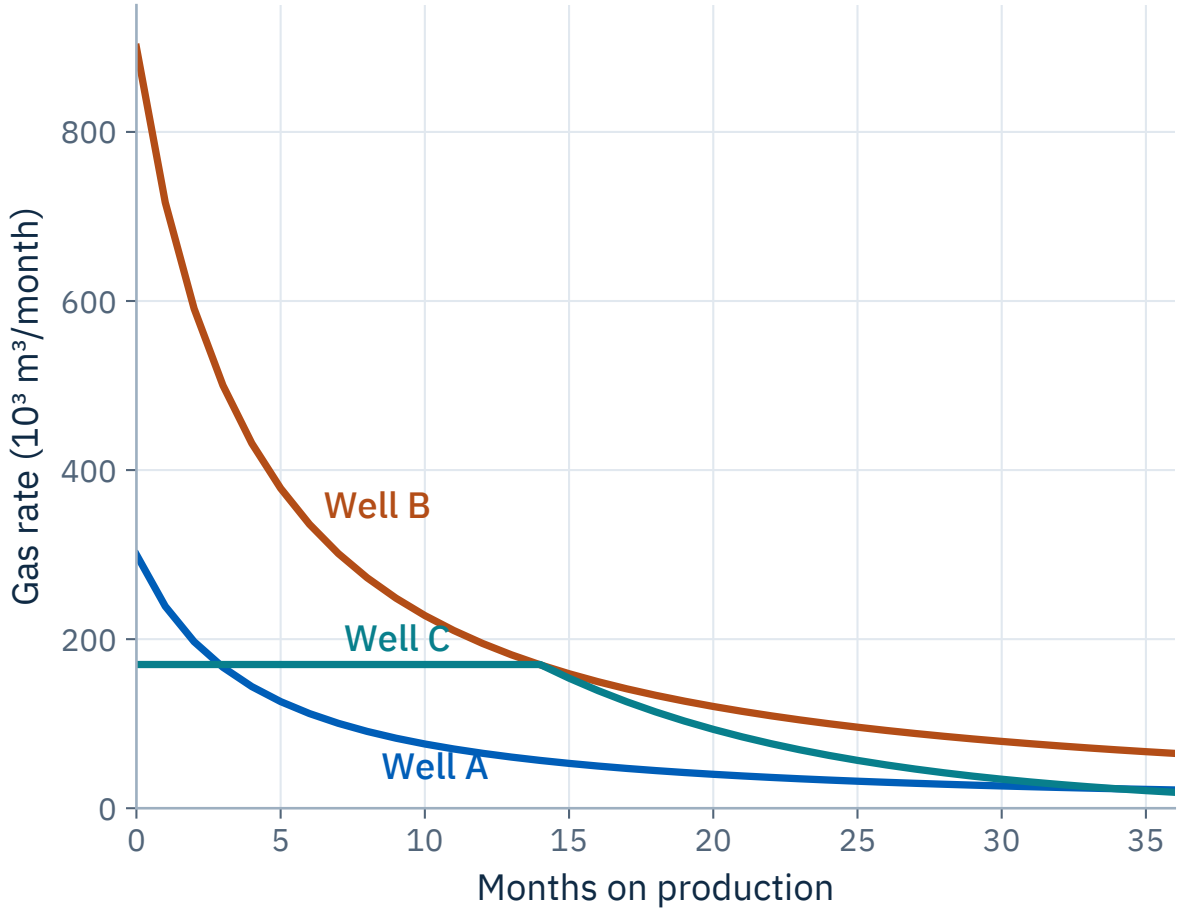
PRACTICE

5 min · Pairs

Similarity is part of the problem definition

You want to group wells with similar production behaviour.

1. Which well is most similar to well A, and why?
2. Should total production level affect similarity?
3. Which variables might simply encode well identity or age?



Synthetic decline curves: B is exactly 3 × A; C has a 14-month plateau.

PART 1 OF 6

01

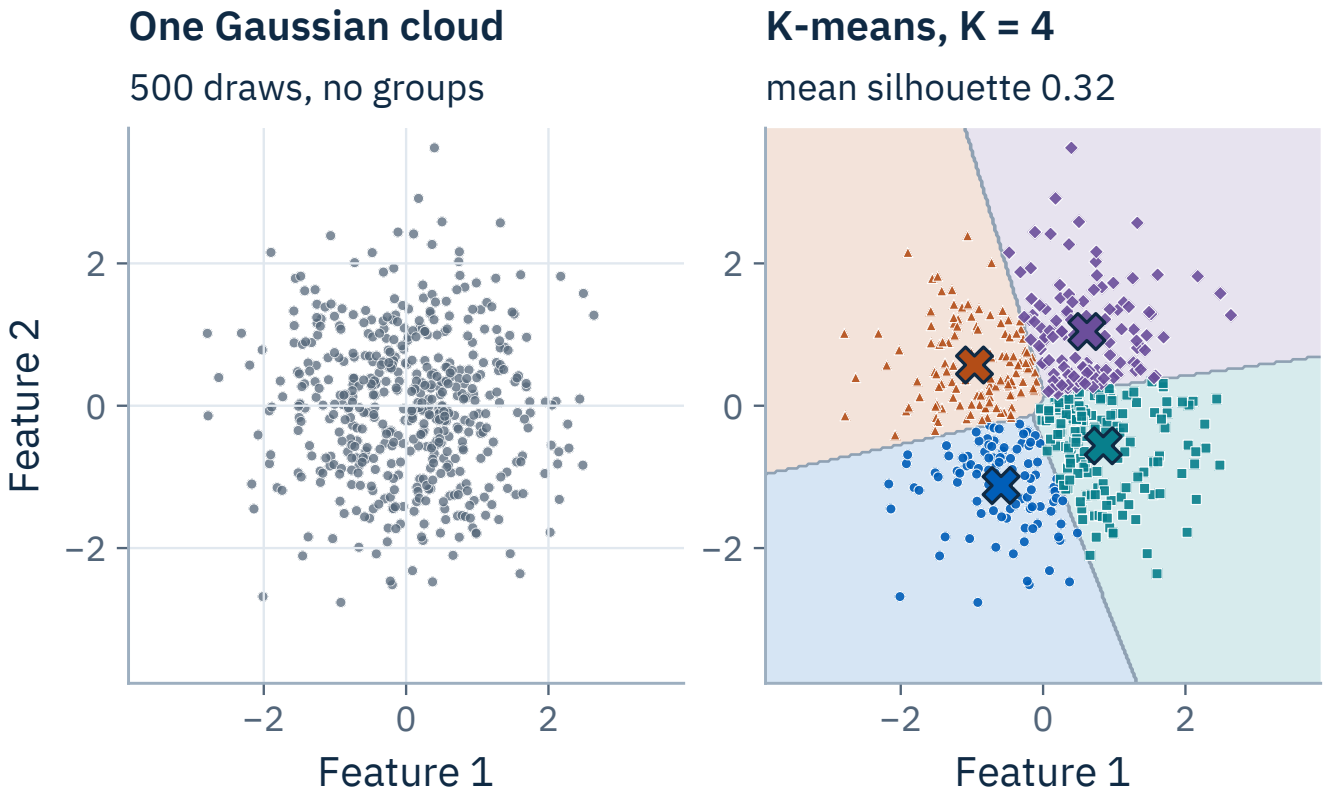
Similarity, features and scaling

A clustering is only as meaningful as its distance.

- Clusters as hypotheses
- The clustering workflow
- Distances, scaling and transforms
- Feature choice and weights

A cluster label is a hypothesis

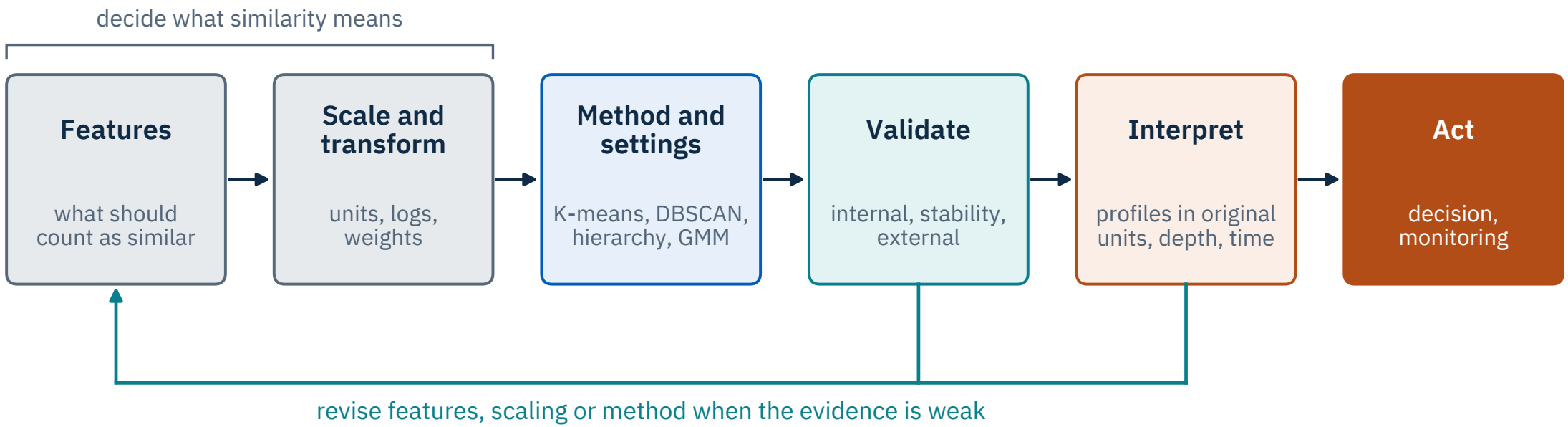
- Algorithms group by their objective and distance, not by physical truth
- Scaling, features and resolution change the groups
- Cluster numbers have no order or physical meaning
- Naming a cluster as a facies needs external evidence



Synthetic: 500 draws from one bivariate normal (seed 11); K-means, K = 4.

Clustering is a loop, not a single fit

Most of the scientific decisions happen before and after the algorithm runs.



Schematic. The feedback loop returns to features, scaling or method whenever validation or interpretation is weak.

The distance decides what counts as similar

Euclidean

- Straight-line distance; K-means uses its square
- Large gaps on one feature dominate

Manhattan

- Sum of absolute differences
- Less dominated by a single large gap

Correlation or cosine

- Compares shape or direction
- Magnitude drops out, e.g. rate level

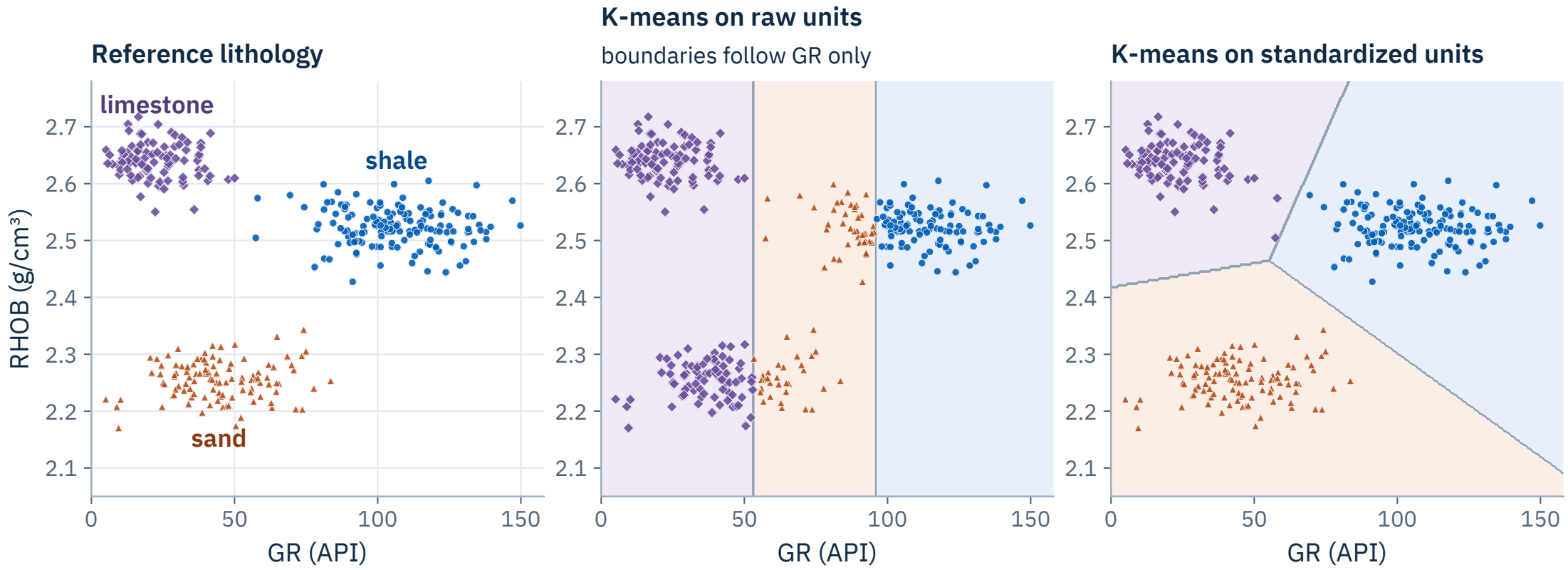
Mahalanobis

- Accounts for spread and correlation
- The distance inside each GMM component

◆ K-means is tied to squared Euclidean distance; DBSCAN and most hierarchical linkages accept other metrics.

Feature scaling changes the clusters

Same data and K: raw GR (SD 39 API) swamps RHOB (SD 0.16 g/cm³) until we rescale.



Synthetic GR and RHOB samples of three lithologies (seed 4); K-means, K = 3, n_init = 10; cluster colours matched to the reference.

Transforms and scalers are modelling choices

Standardize

- Mean 0 and SD 1 per feature
- Equal variance, not equal importance

Robust scale

- Median and interquartile range
- Outliers no longer set the scale

Log first

- Resistivity, permeability, rates
- Skewed, multiplicative variables

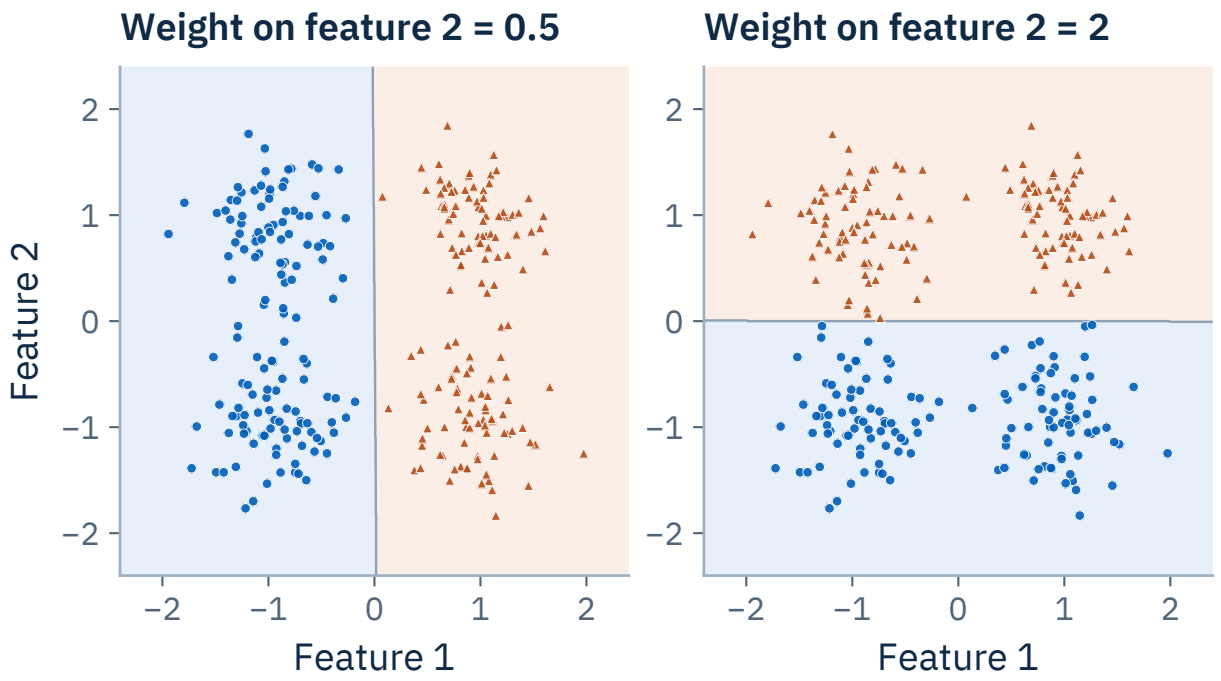
Keep units

- When features share one unit and scale
- e.g. normalized production profiles

◆ Fit the scaler once on the clustering data and reuse the fitted object for every new well.

Feature choice and weights set the answer

- Each feature's influence grows with its spread after scaling
- A duplicated or strongly correlated log counts twice
- Depth, well ID and dates encode location, not rock or behaviour
- Weight deliberately: multiply a standardized column by w



Synthetic four-group data (seed 8), standardized; K-means, K = 2, with feature 2 multiplied by the stated weight.

PRACTICE

5 min · Pairs

Choose the features for an electrofacies study

Available for 12 wells: GR, RHOB, NPHI, RT, DT, CALI, DEPTH and well name.

1. Which logs would you cluster on, and which would you leave out?
2. Which need a transform before scaling?
3. What would you check before trusting the clusters?

PART 2 OF 6

02

K-means

A compact-cluster model based on squared Euclidean distance

- The objective and Lloyd iterations
- Initialization: k-means++ and restarts
- Choosing K with elbow and silhouette
- Failure modes and variants

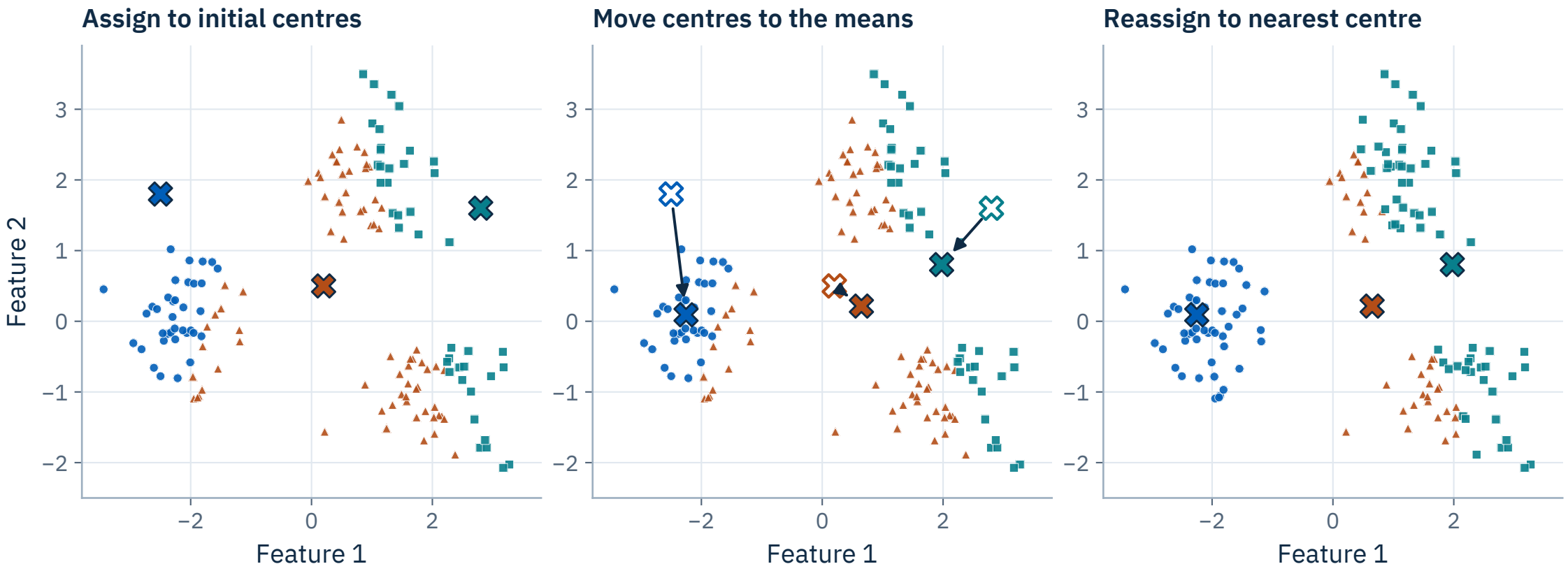
K-means minimizes within-cluster squared distance

$$J = \sum_{i=1}^{n} \| x_i - \mu_{c_i} \|^2$$

1. Choose K initial centroids
2. Assign each observation to its nearest centroid
3. Move each centroid to the mean of its assigned observations
4. Repeat until the assignments stop changing

- Neither step can increase J, so the loop converges, but only to a local minimum.

Assignment and update are separate steps



Synthetic three blobs (random_state = 42); one Lloyd iteration from fixed starting centres. Crosses mark centroids (hollow: starting position); colours show current assignments.

PRACTICE

4 min · Individual

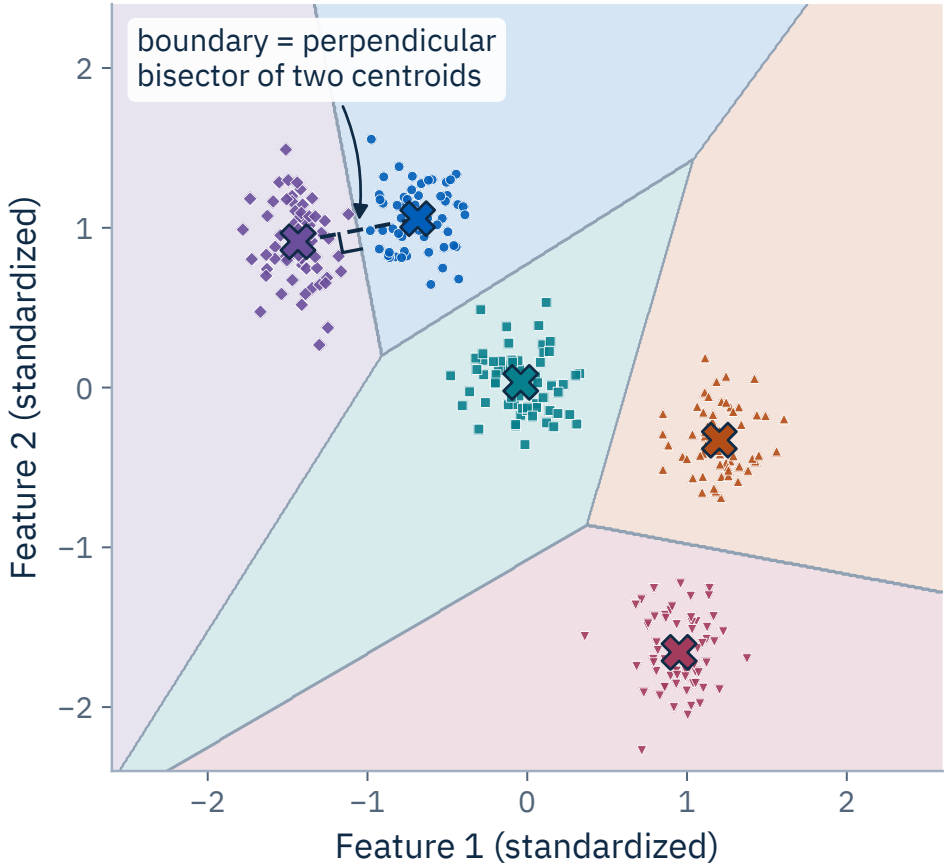
One update by hand

Observations on a line: 1, 2, 3, 8, 9, 10. Initial centroids: 2 and 8.

1. Assign each observation to a centroid.
2. Calculate the updated centroids.
3. Calculate the within-cluster sum of squares before and after the update.

Assignments carve the space into Voronoi cells

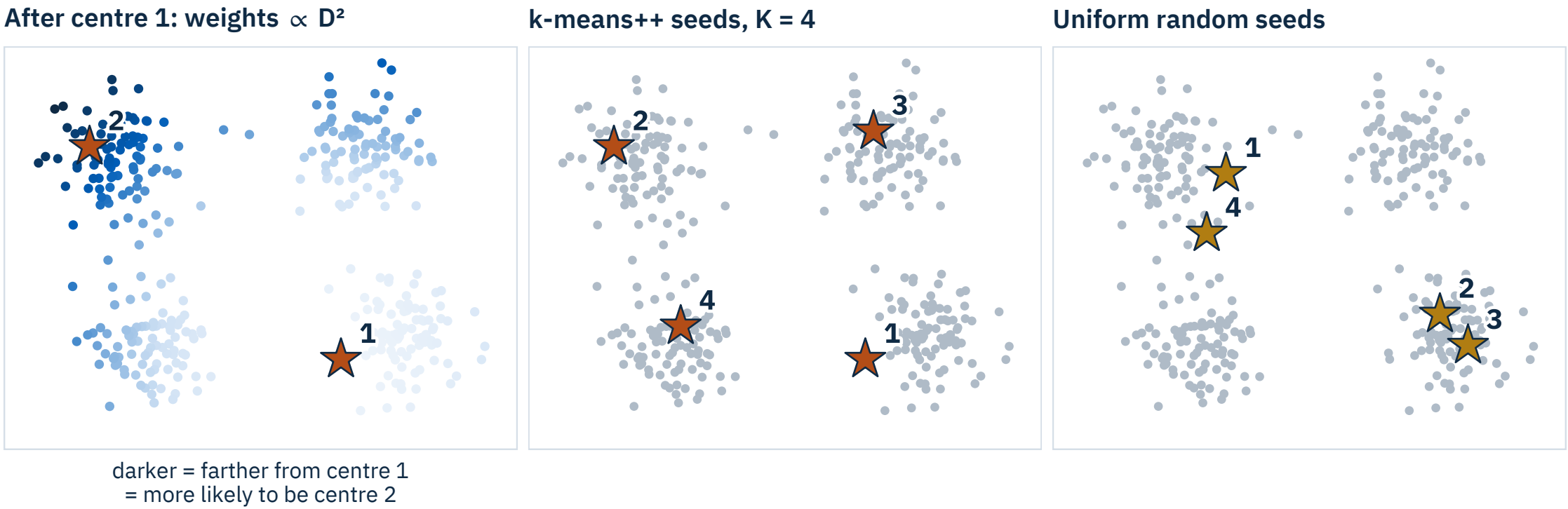
- Each point joins its nearest centroid
- Boundaries are perpendicular bisectors between centroids
- Every cell is convex: one cluster cannot curve around another
- New samples are assigned by the same rule with `predict`



Synthetic blobs (random_state = 6), standardized; K-means, K = 5. Shading marks each centroid's cell.

k-means++ spreads the initial centroids

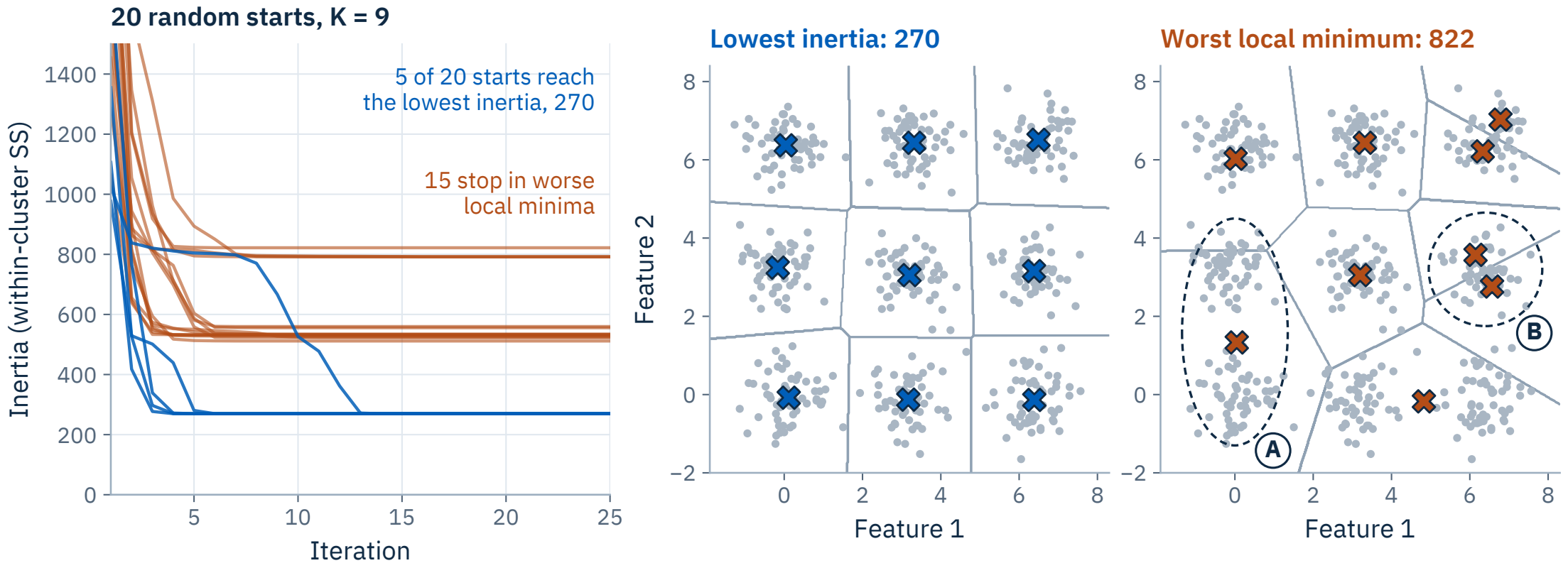
Each new centre is sampled with weight D^2: squared distance to the nearest centre.



Synthetic four blobs (random_state = 2); one draw each. One seed per blob in 66% of 2,000 textbook k-means++ draws, 95% with scikit-learn's greedy variant, 9.5% for uniform sampling (exact).

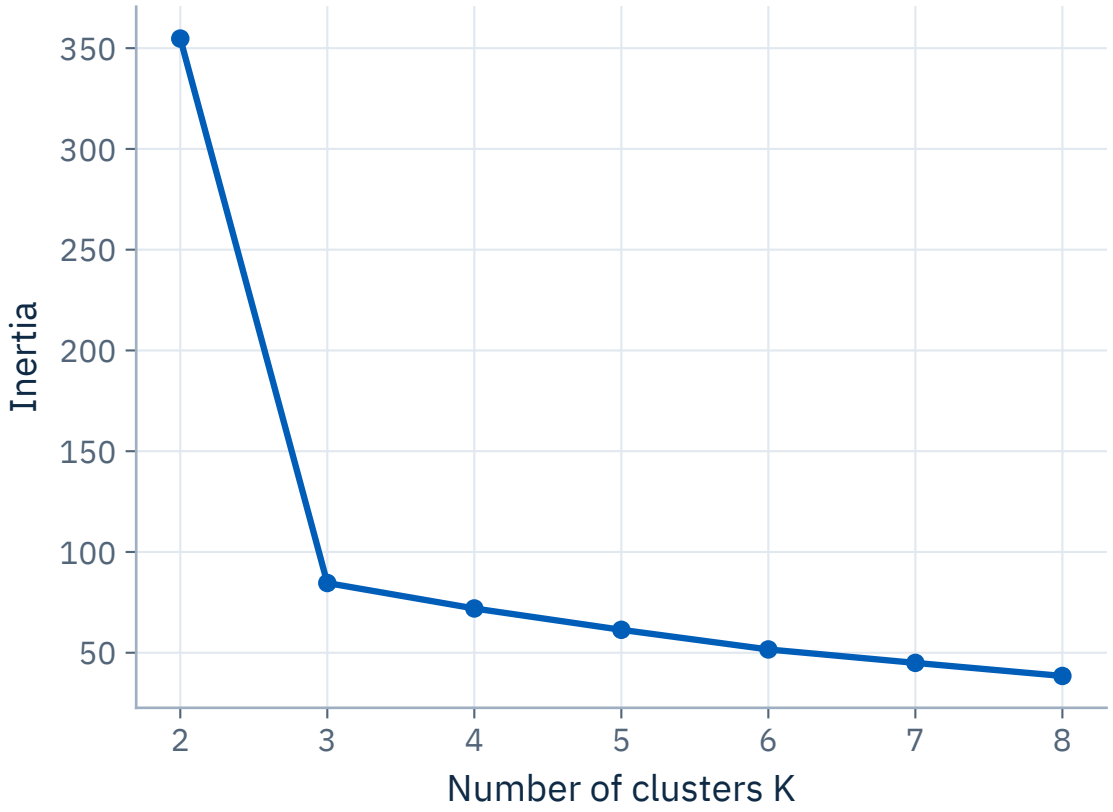
Random starts end in different local minima

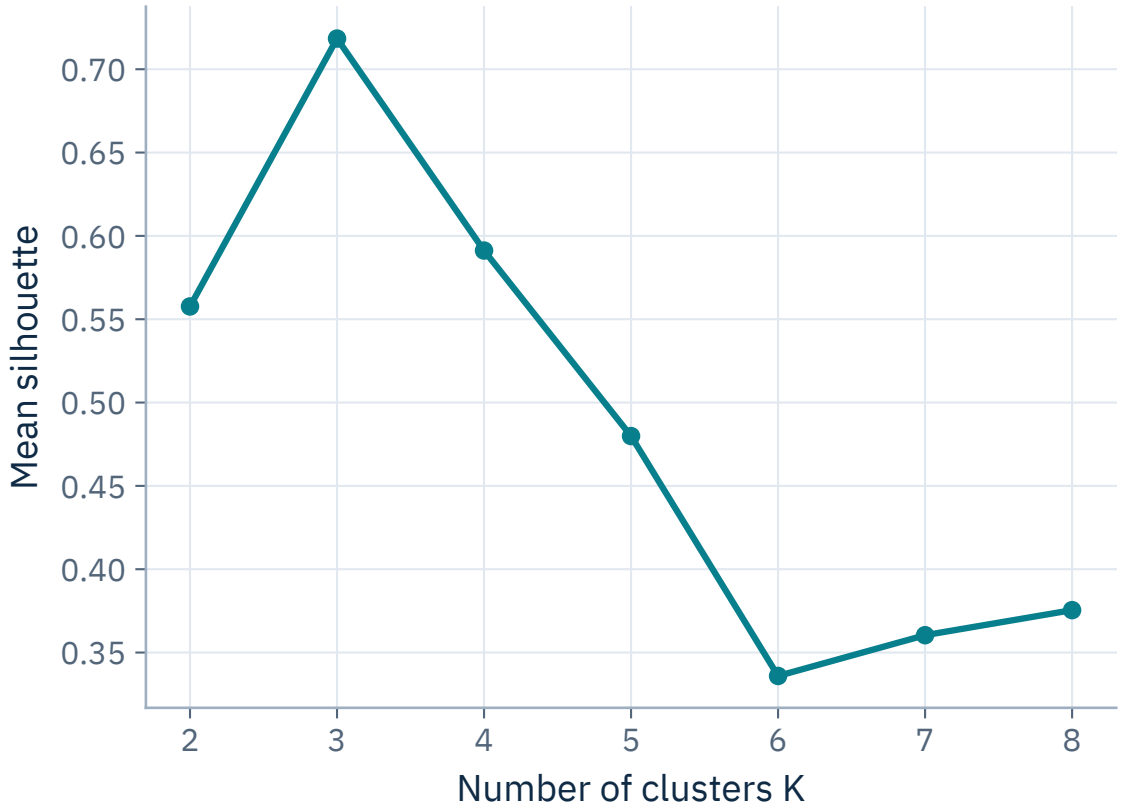
Lloyd iterations only go downhill; where they stop depends on where they start.



Synthetic 3 × 3 blob grid; 20 uniform random starts, K = 9. A: one centroid covers two blobs. B: two centroids split one blob.

Choosing K needs more than an elbow





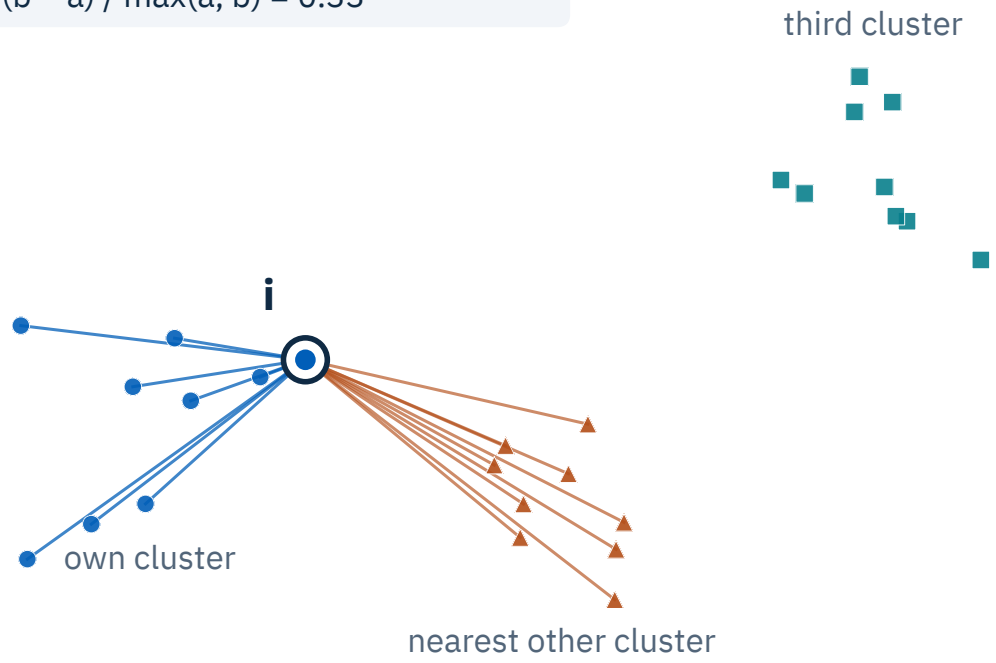
Inertia and mean silhouette measured on one synthetic dataset. Both depend on the feature space and cluster assumptions.

Silhouette compares cohesion with separation

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$

- a_i: mean distance to the other members of its own cluster
- b_i: smallest mean distance to the members of another cluster
- Near 1: well placed; near 0: on a boundary; negative: closer to another cluster

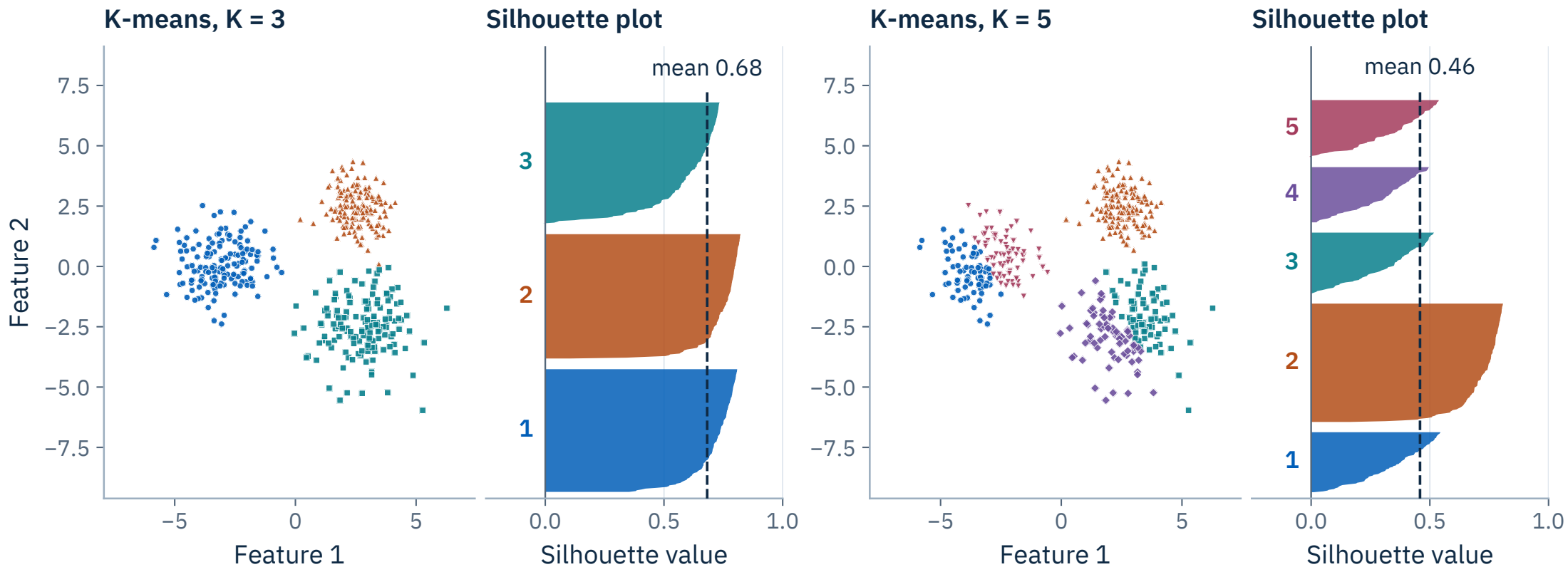
a = 1.53 mean distance to own cluster
b = 2.27 mean distance to nearest other
s = (b − a) / max(a, b) = 0.33



Synthetic three-cluster example (seed 21); values checked against scikit-learn silhouette_samples.

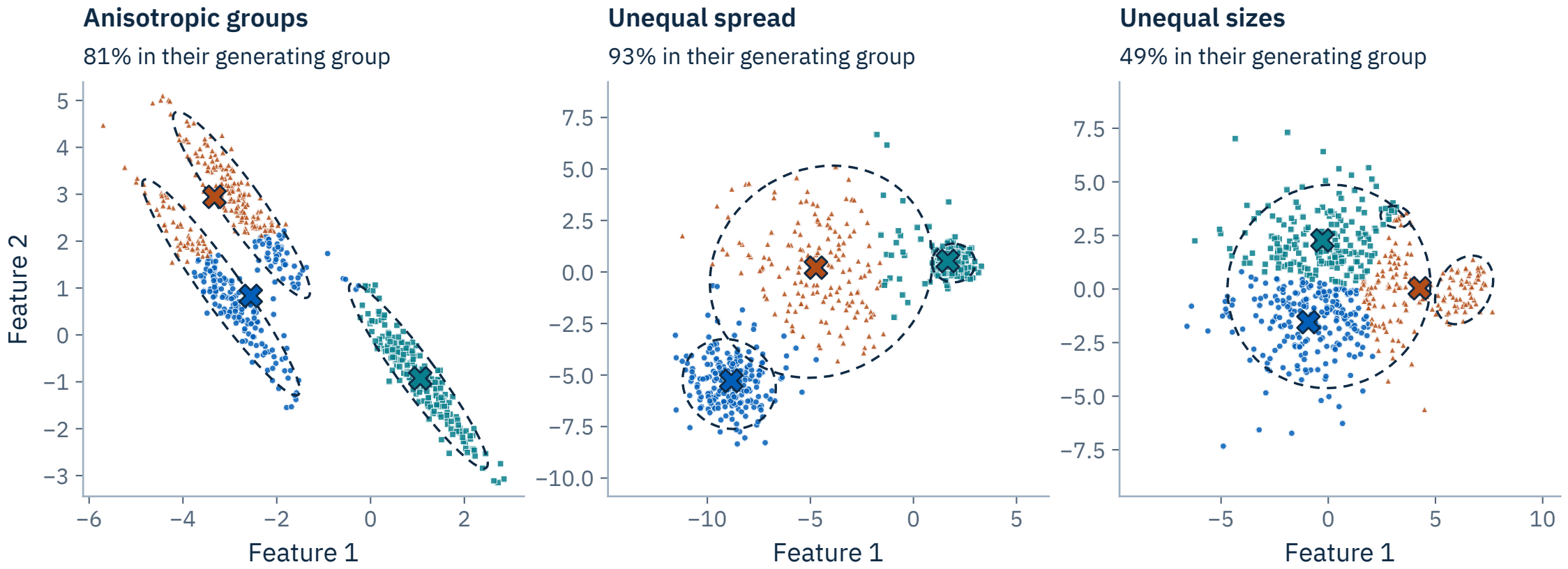
A silhouette plot shows every observation

Read each knife's width (size), length (fit) and tail (low or negative values).



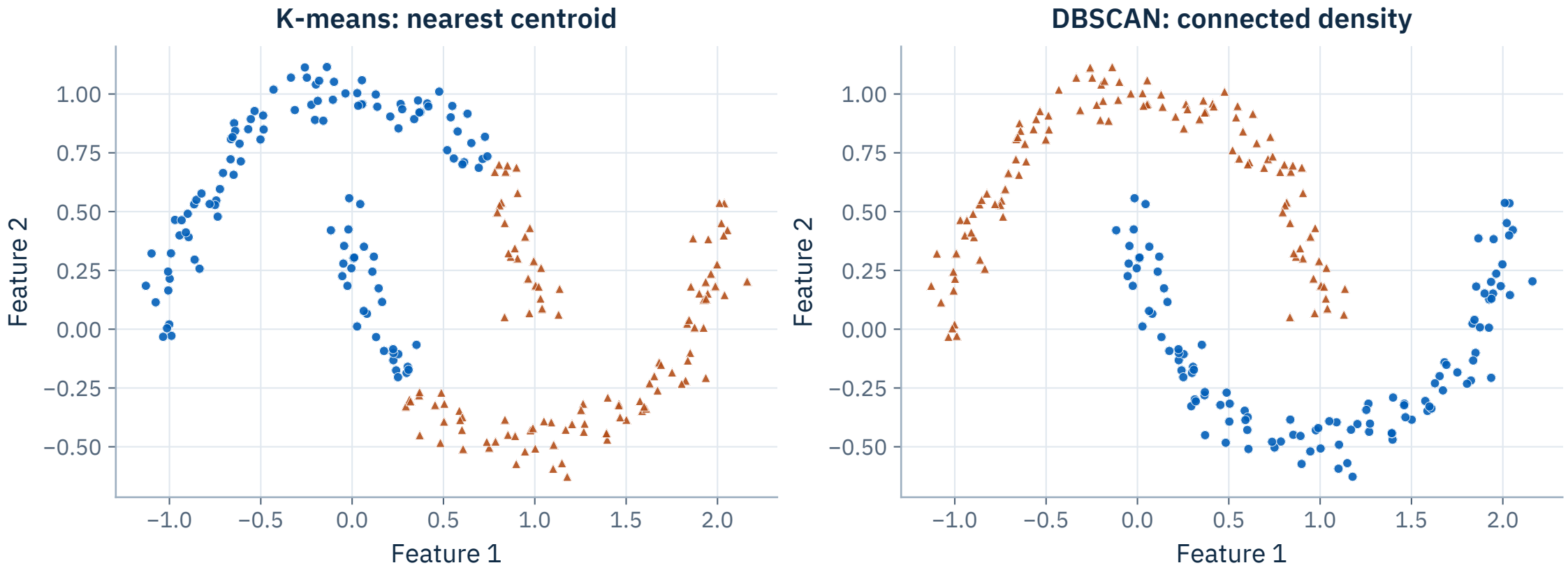
Synthetic three-blob data (random_state = 5); K-means with K = 3 and K = 5; per-sample values from silhouette_samples.

K-means fails in predictable ways



Synthetic data after the scikit-learn k-means assumptions example; K-means, K = 3, n_init = 10. Dashed ellipses: generating groups (90%).

The same data, different assumptions



K-means and DBSCAN fitted to the same synthetic two-moon data. Grey crosses mark density noise when present.

K-means works best with compatible geometry

✓ Works well when

- Groups are compact and similar in spread
- Features are scaled to comparable influence
- K can be checked with several kinds of evidence

vs

✗ Struggles when

- Groups are elongated, curved or nested
- Densities or sizes differ strongly
- Outliers pull centroids away

◆ Standardization is a choice: it can raise a noisy low-variance log to full weight.

Variants trade exactness for speed or robustness

MiniBatchKMeans

- Updates centroids from small random batches
- Much faster on large n; quality slightly reduced

k-medoids

- Centres are actual observations
- Any distance; less pulled by outliers
- Not in scikit-learn itself (scikit-learn-extra)

BisectingKMeans

- Splits one cluster at a time
- Produces a hierarchy of K-means splits

◆ Use MiniBatchKMeans when a full fit is too costly, and compare it with KMeans on a subsample.

A reproducible K-means fit

```
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

model = make_pipeline(
    StandardScaler(),
    KMeans(n_clusters=4, init="k-means++", n_init=10, random_state=0),
)
labels = model.fit_predict(logs[["GR", "RHOB", "NPHI", "LOG_RT"]])
km = model[-1]
centres = model[0].inverse_transform(km.cluster_centers_)  # original units
print(km.inertia_, km.n_iter_)
```

Since scikit-learn 1.4, n_init="auto" runs k-means++ once: set n_init yourself.

PRACTICE

5 min · Pairs

How many restarts are enough?

In the random-start experiment, 5 of 20 starts reached the lowest inertia.

1. Estimate p, the chance that one random start succeeds.
2. What is the chance that n_init = 10 finds the best solution at least once?
3. How many starts give 99%, and what are you assuming?

LABORATORY

50 min

Lab 1: K-means electrofacies on well W5

- ☐ Load w5.csv, inspect the six logs and standardize them.
- ☐ Fit K-means with K = 3 and n_init = 10; plot crossplots by cluster.
- ☐ Compare K = 2–6 with inertia, mean silhouette and silhouette plots.
- ☐ Profile the chosen clusters in original units and plot them against depth.

NOTEBOOK

`exercises/01_kmeans/KMeans_Clustering_Student.ipynb`

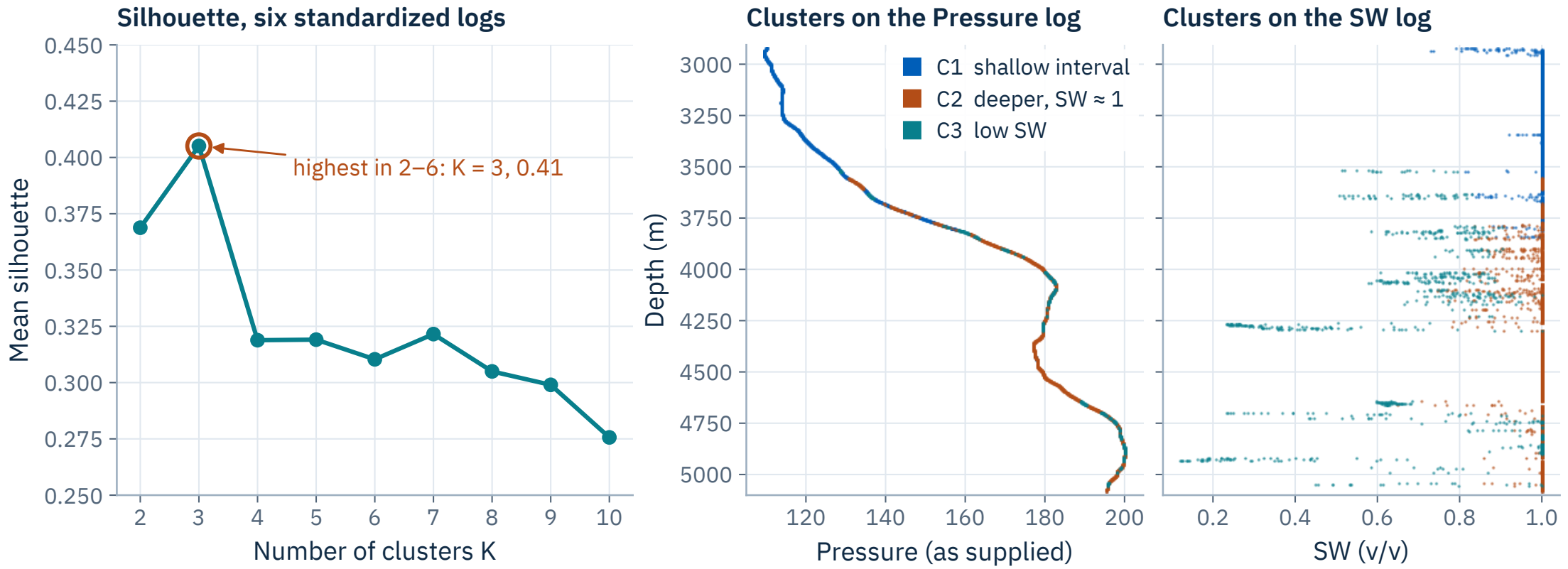
EVIDENCE TO SUBMIT

- Elbow and silhouette plots with your chosen K
- A cluster profile table in original units
- A depth plot and one hypothesis per cluster

DISCUSSION

Lab 1 debrief: W5 clusters follow depth and SW

The best silhouette (0.41, K = 3) mostly splits depth and low SW, not lithology.



Computed from w5.csv (course data): six standardized logs, KMeans(n_init = 10, random_state = 42), as in the notebook.

PART 3 OF 6

03

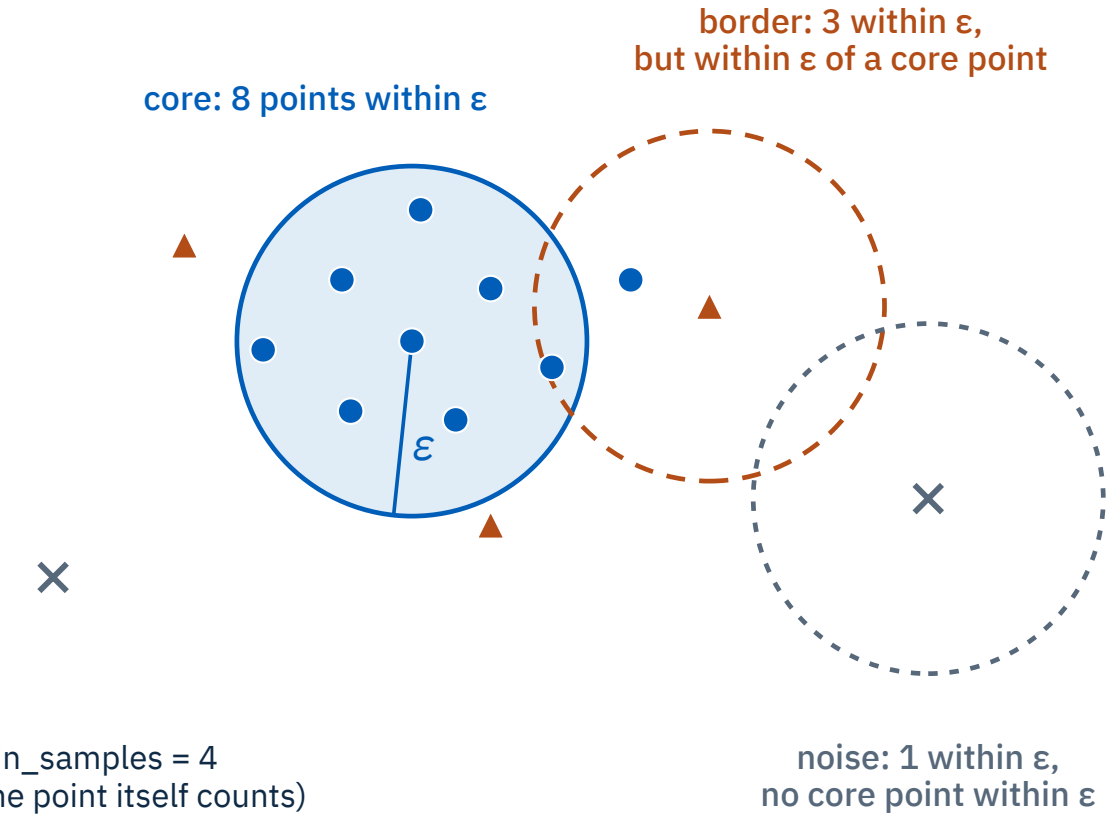
Density-based clustering

Clusters are dense regions separated by sparse ones.

- Core, border and noise points
- Choosing eps with a k-distance plot
- HDBSCAN for varying densities
- OPTICS reachability plots

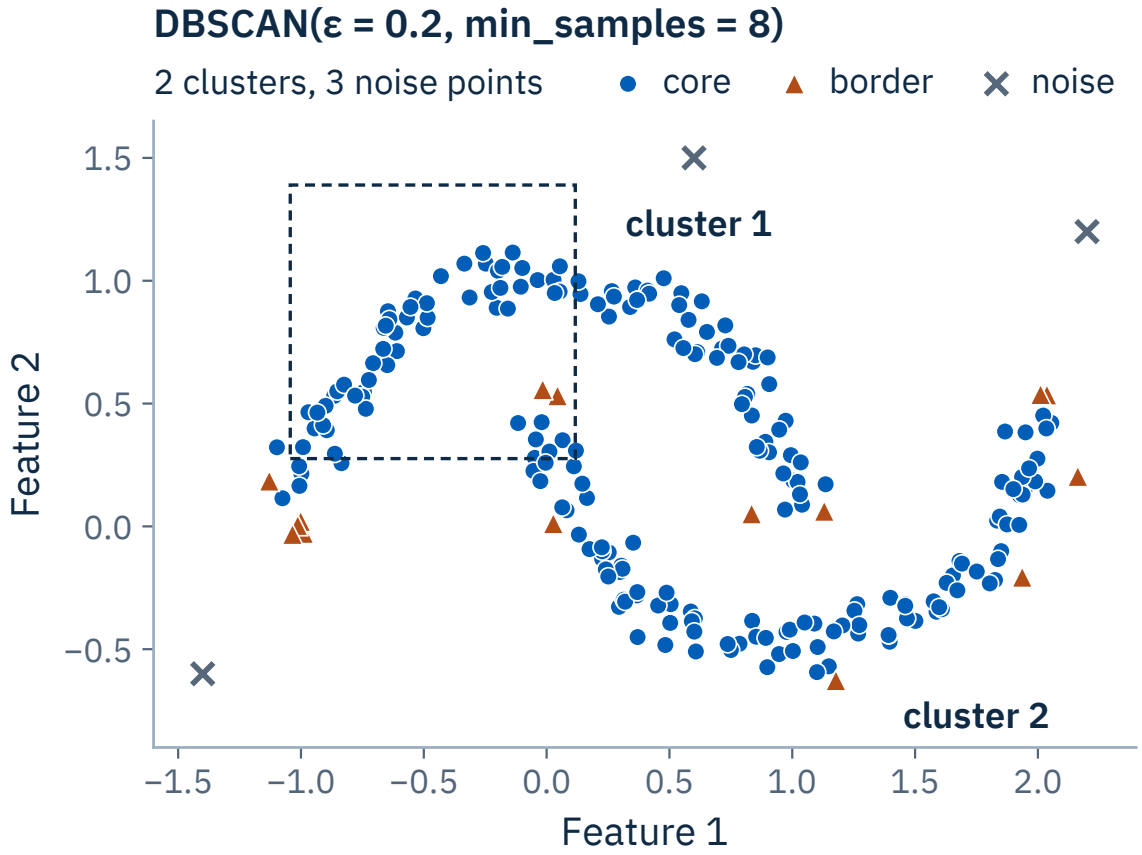
DBSCAN labels points as core, border or noise

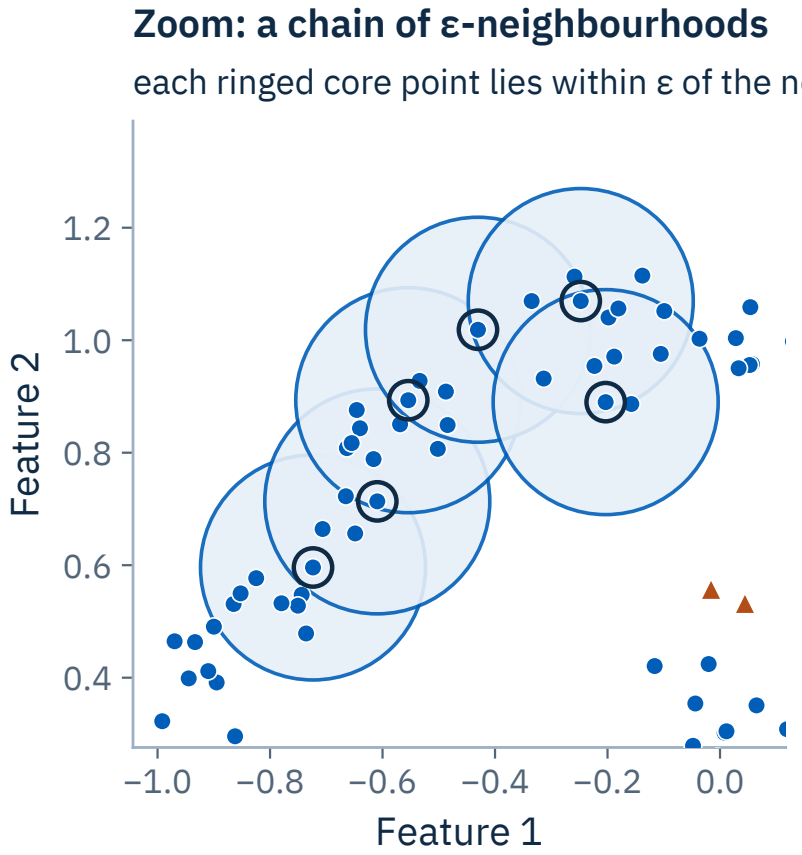
- **Core**: at least min_samples points within ε, counting itself
- Border: not core, but within ε of a core point
- Noise: neither; label −1
- Clusters are core points linked through overlapping ε-neighbourhoods



Hand-placed points; types computed with DBSCAN(eps = 1.0, min_samples = 4).

Clusters grow by chaining core neighbourhoods





Synthetic two moons (random_state = 42) plus three outliers; DBSCAN(eps = 0.2, min_samples = 8). Right: ε-discs along a chain of core points.

eps and min_samples define a density scale

eps

- Neighbourhood radius in the scaled feature space
- Too small: most points become noise
- Too large: distinct groups merge

min_samples

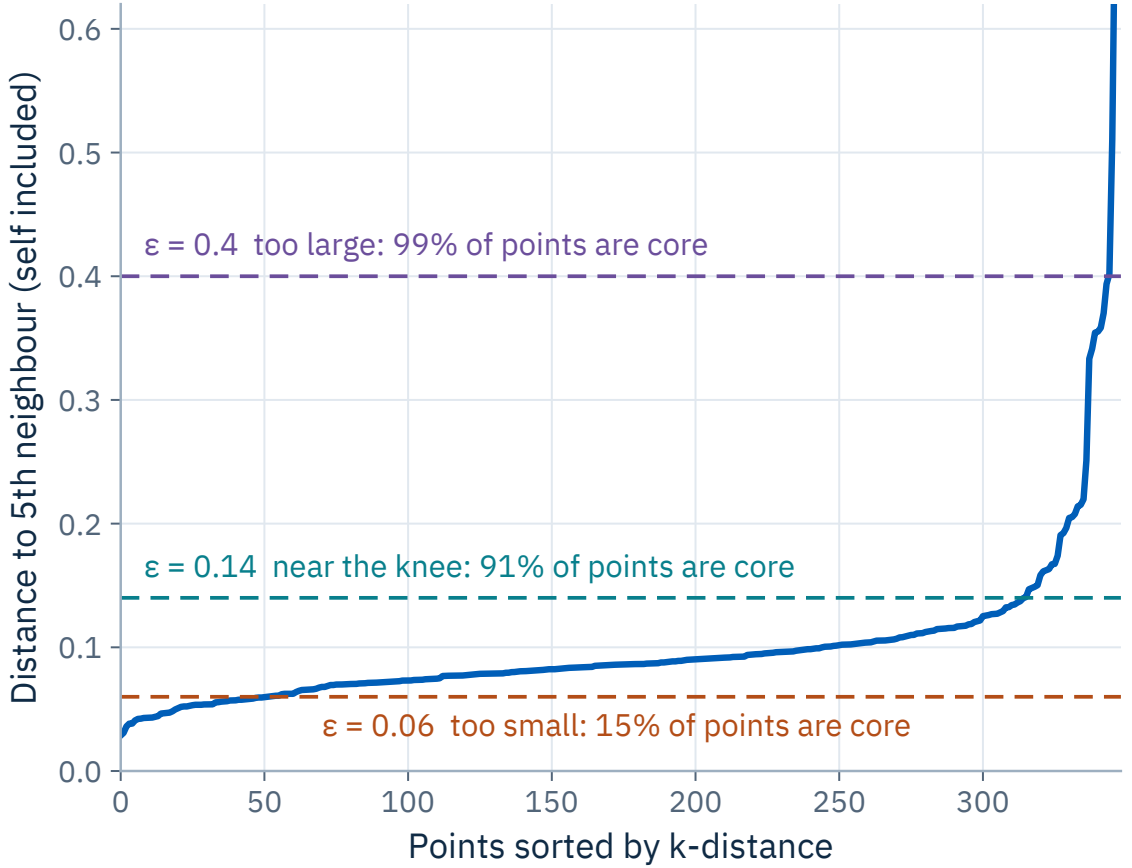
- Points required within eps, including itself
- Larger values: smoother clusters, more noise
- Heuristic start: about 2 × number of features

Together

- Set a minimum density for a core point
- Re-tune after any change of features or scaling
- Report the noise fraction with every result

A k-distance plot suggests a range for eps

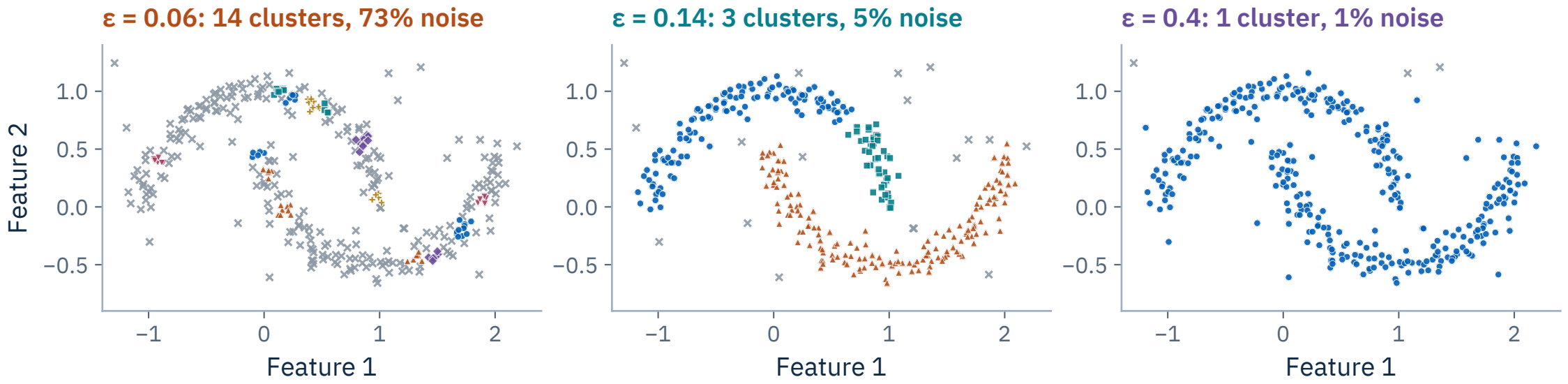
- For each point, find the distance to its min_samples-th neighbour
- Sort and plot these k-distances
- Points below the chosen eps are core points
- Pick eps near the knee, then check a range around it



Synthetic moons (random_state = 3) plus 28 uniform noise points; distances to the 5th neighbour, counting the point itself.

Small changes in eps change the answer

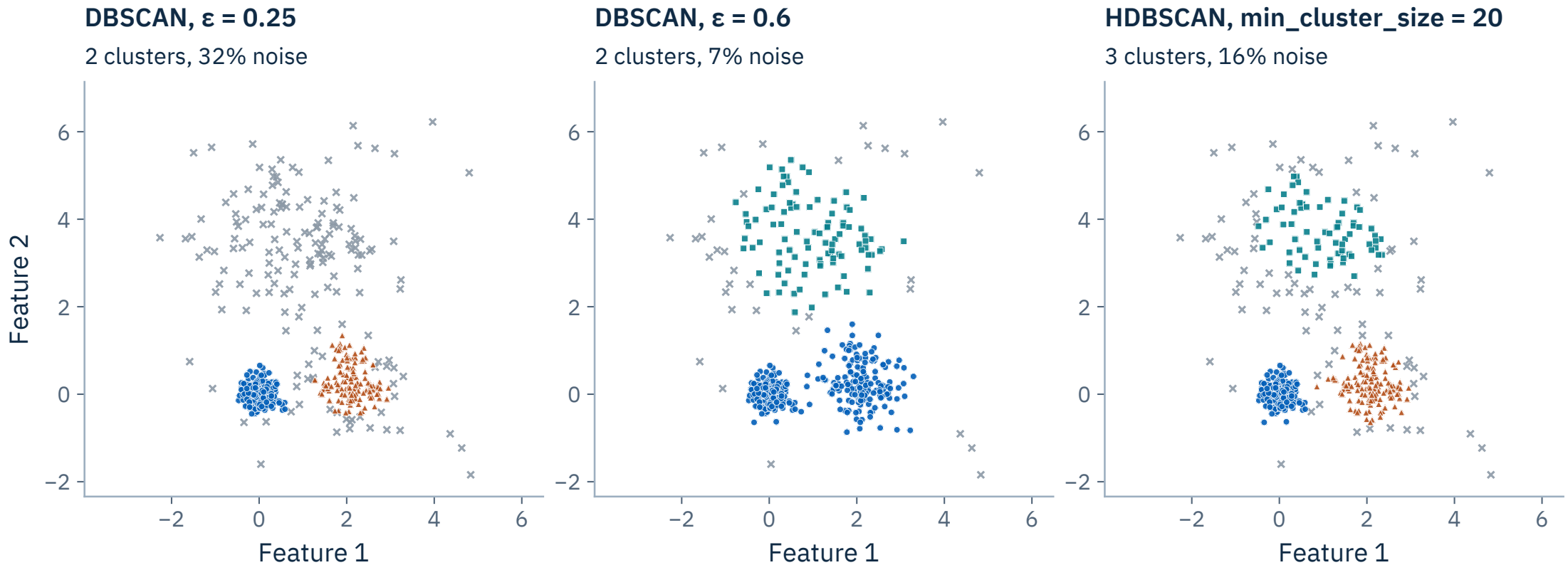
The knee value recovers both moons; either side of it DBSCAN fragments or merges them.



Same synthetic data as the k-distance plot; DBSCAN(min_samples = 5) at three eps values. Grey crosses are noise.

One eps cannot fit clusters of different density

Small eps loses the sparse group; large eps merges the dense ones. HDBSCAN adapts.



Synthetic dense, medium and sparse groups plus uniform background (seed 14); DBSCAN(min_samples = 10) and HDBSCAN(min_cluster_size = 20).

HDBSCAN keeps the clusters that persist across densities

HDBSCAN runs DBSCAN at every density level at once and keeps the clusters that persist.

1

Core distance

distance to the min_samples-th neighbour

2

Mutual reachability

max of both core distances and their distance

3

Spanning tree

connects all points at lowest total distance

4

Hierarchy

cut tree edges from longest to shortest

5

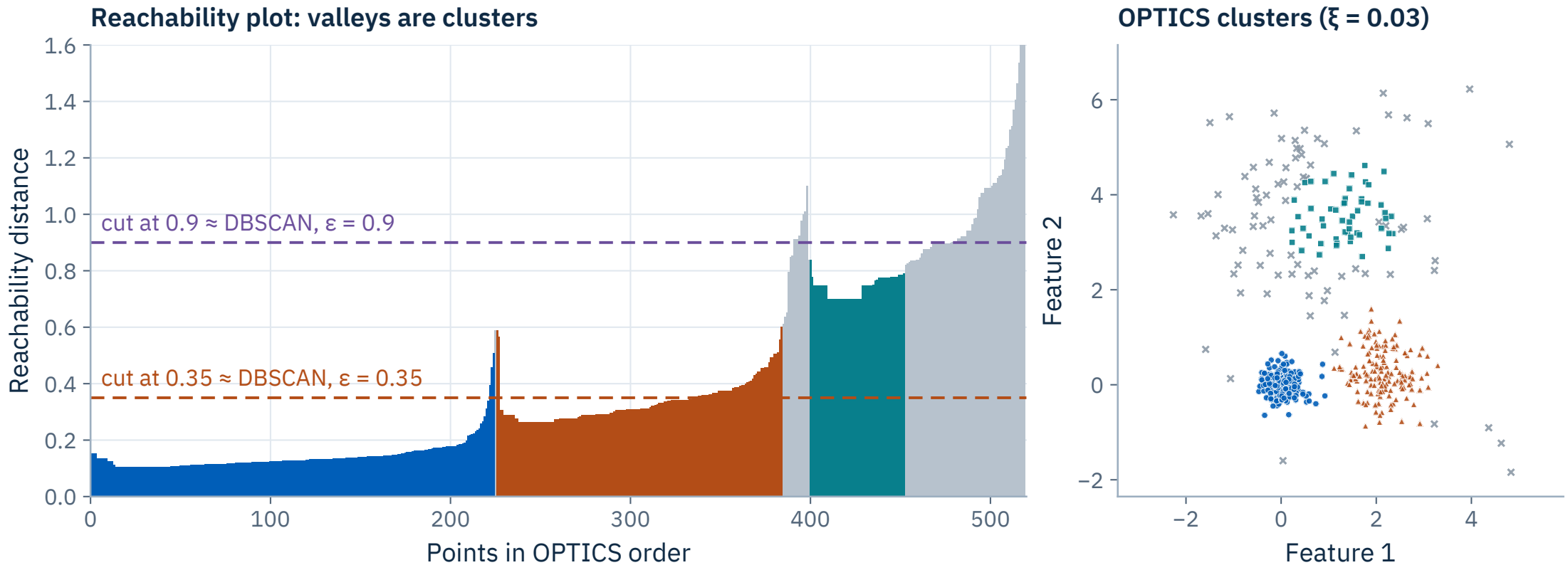
Stable clusters

keep clusters that persist; smaller than min_cluster_size is noise

◆ HDBSCAN needs no eps; min_cluster_size is the main setting.

OPTICS orders points by reachability

Valleys are dense regions; a cut at height ε reproduces DBSCAN with that ε.



Same synthetic data as the HDBSCAN comparison; OPTICS(min_samples = 30, xi = 0.03). Bars are coloured by extracted cluster; grey = noise.

HDBSCAN and OPTICS adapt where DBSCAN cannot

Method	Main settings	Varying density	New samples
DBSCAN	eps, min_samples	Poor: one global eps	No predict: refit
HDBSCAN	min_cluster_size	Good	No predict in scikit-learn
OPTICS	min_samples, xi	Good	No predict: refit
K-means (reference)	K	Poor	predict: nearest centroid

All three density methods label noise as −1 and need a scaled, meaningful distance.

Tune eps from k-distances, then report the noise

```
from sklearn.neighbors import NearestNeighbors
from sklearn.cluster import DBSCAN

ms = 5                                          # counts the point itself
nn = NearestNeighbors(n_neighbors=ms).fit(Z)    # Z: scaled features
kdist = np.sort(nn.kneighbors(Z)[0][:, -1])     # plot; look for a knee

db = DBSCAN(eps=0.5, min_samples=ms).fit(Z)
labels = db.labels_                             # -1 marks noise
n_clusters = len(set(labels) - {-1})
noise_frac = np.mean(labels == -1)
print(n_clusters, round(noise_frac, 3))
```

Report the number of clusters, the noise fraction and the settings together.

PRACTICE

5 min · Pairs

Density clustering diagnosis

DBSCAN assigns 85% of observations to noise.

1. List two possible explanations other than bad data.
2. What would you plot or inspect before changing eps?
3. What happens if you simply raise eps until noise is below 5%?

LABORATORY

45 min

Lab 2: density clustering on log data

- Standardize SP and GR from log.csv; run DBSCAN with eps = 0.5, min_samples = 5.
- Draw the k-distance plot and mark the default eps on it.
- Sweep eps and min_samples; tabulate clusters, noise and largest cluster.
- Repeat on the six W5 logs and inspect what the noise points have in common.

NOTEBOOK

`exercises/02_dbscan/DBSCAN_Clustering_Student.ipynb`

EVIDENCE TO SUBMIT

- A k-distance plot with your chosen eps
- A sensitivity table for eps and min_samples
- Two sentences on what the noise represents

DISCUSSION

Lab 2 debrief: noise flags rare responses, not errors

eps (standardized)	log.csv: clusters, noise	W5: clusters, noise	W5 largest cluster
0.3	5, 57 of 110	76, 30%	1,699
0.5	1, 8 of 110	15, 6.6%	6,342
0.8	1, 4 of 110	3, 1.2%	6,969
1.0	1, 1 of 110	3, 0.3%	7,037

With min_samples = 5, both datasets are essentially one dense region plus sparse outliers.

PART 4 OF 6

04

Hierarchy and zonation

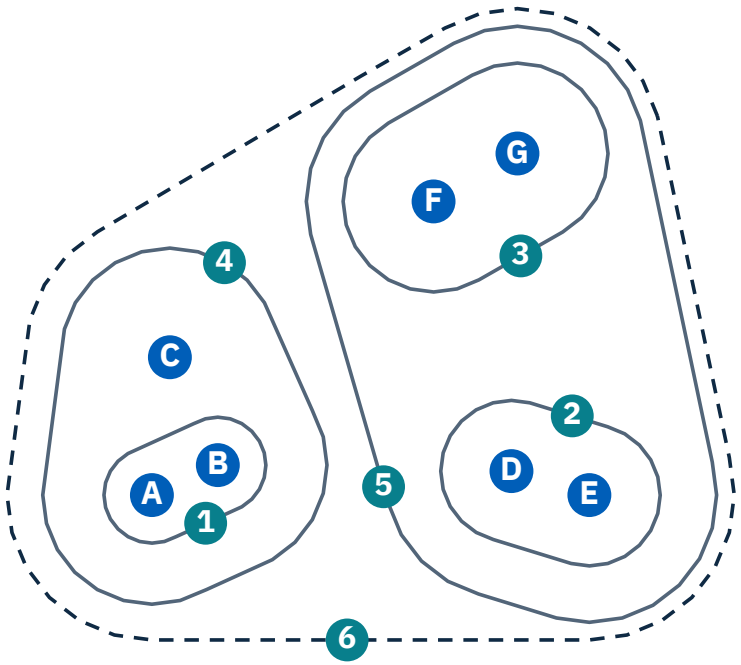
Nested merges give every level of detail at once.

- Agglomerative clustering and linkage
- Reading and cutting dendrograms
- Depth-constrained zonation of a well
- Cost and scaling limits

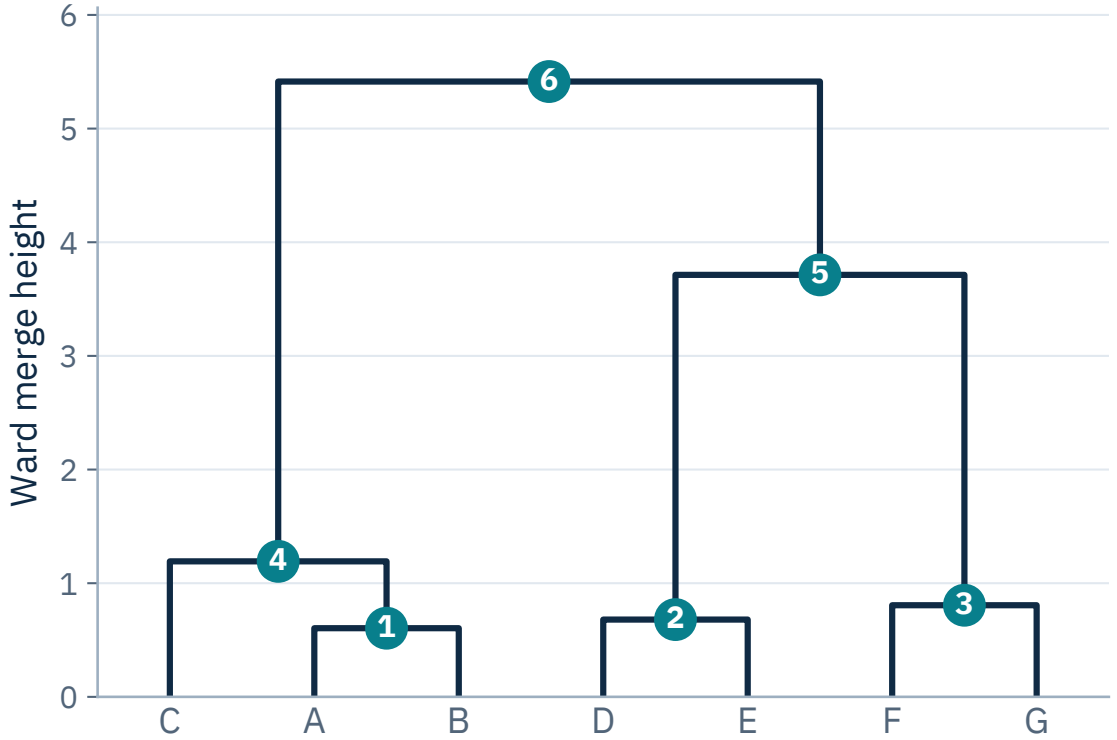
Agglomerative clustering builds a merge history

Start with singletons, merge the closest pair of clusters, and record each height.

Nested merges (numbers = merge order)



Dendrogram of the same merges



Seven hand-placed points; scipy Ward linkage. Merge numbers follow the rows of the linkage matrix.

Linkage changes the meaning of close

Linkage	Distance between groups	Common behaviour
Single	Closest pair of observations	Can chain through bridges
Complete	Farthest pair	Favours compact groups
Average	Mean of all pairwise distances	Intermediate behaviour
Ward	Increase in within-cluster squared error	Compact, similar-size groups

Ward is defined for Euclidean distance; the other three accept any distance.

Four linkages measure the same two groups differently

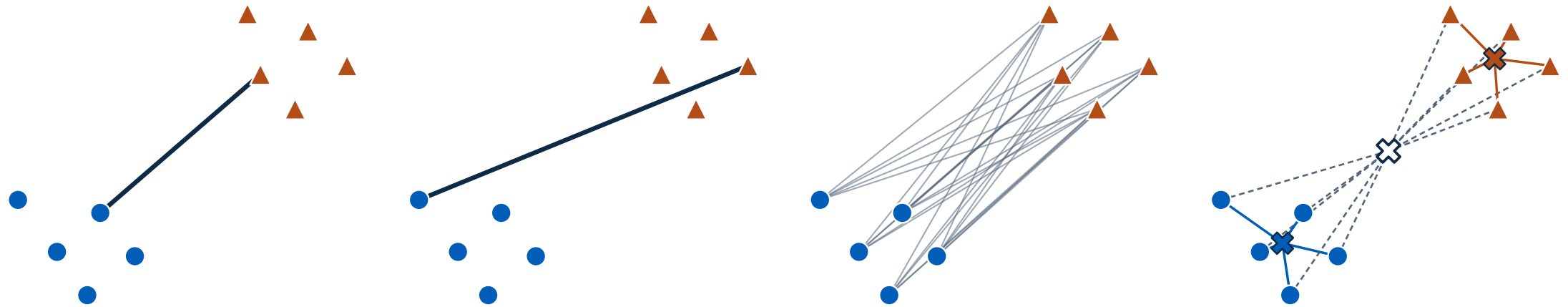
Same two groups, four definitions of the distance between them.

Single: closest pair
d = 2.45

Complete: farthest pair
d = 4.10

Average: mean of all 25 pairs
d = 3.31

Ward: growth in within-SS
ΔSS = 26.35



Two hand-placed five-point groups. Ward: increase in within-cluster sum of squares if the groups merge (dashed = distances to the merged centroid).

PRACTICE

5 min · Individual

Merge heights by hand

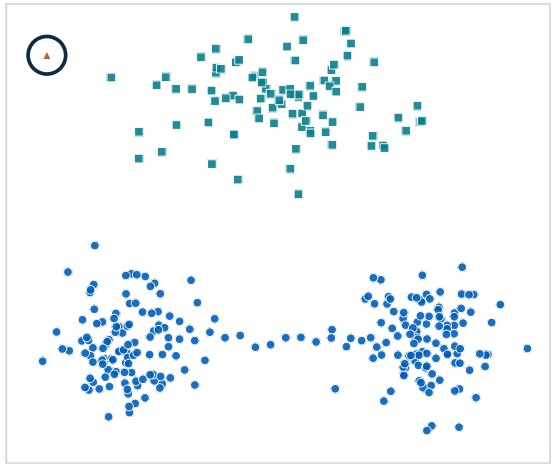
Five observations on a line: 0, 1, 3, 7, 8.

1. List the merges and their heights with single linkage.
2. Repeat with complete linkage.
3. Where would you cut each tree to get two clusters?

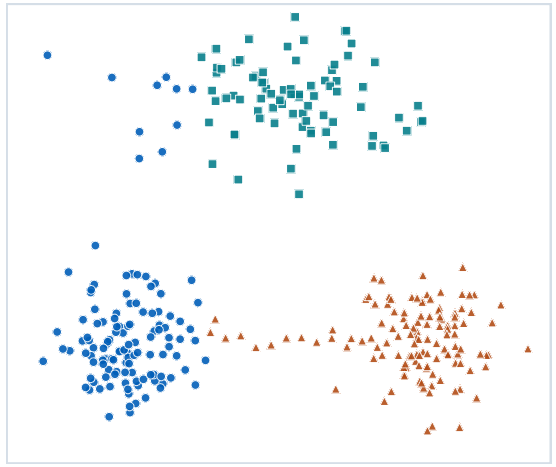
Single linkage chains through bridges

Single linkage merges two blobs through a 14-point bridge and isolates one point.

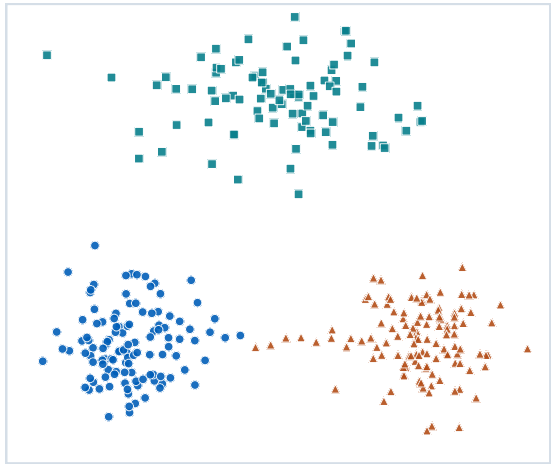
Single
sizes 234 · 89 · 1 (ringed)



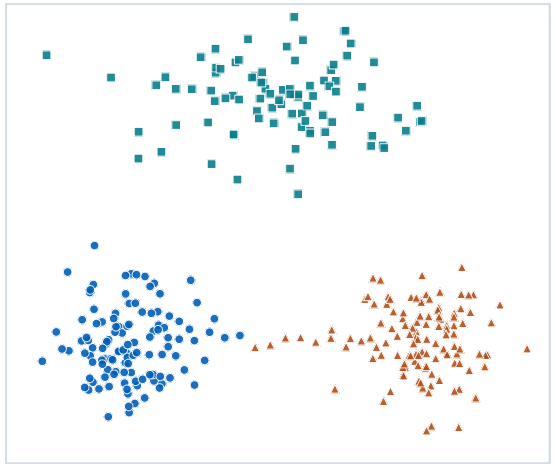
Complete
sizes 123 · 121 · 80



Average
sizes 119 · 115 · 90

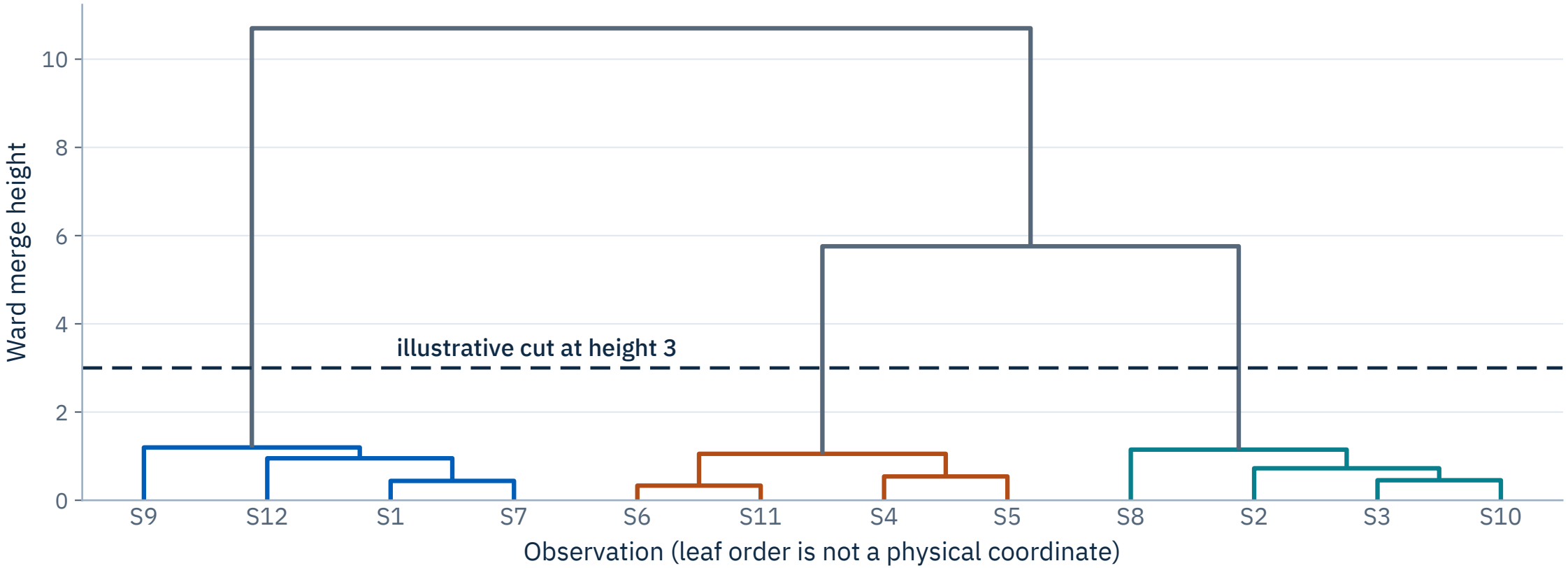


Ward
sizes 119 · 115 · 90



Synthetic: two dense blobs joined by a thin bridge, plus an elongated third group (seed 5); AgglomerativeClustering(n_clusters = 3).

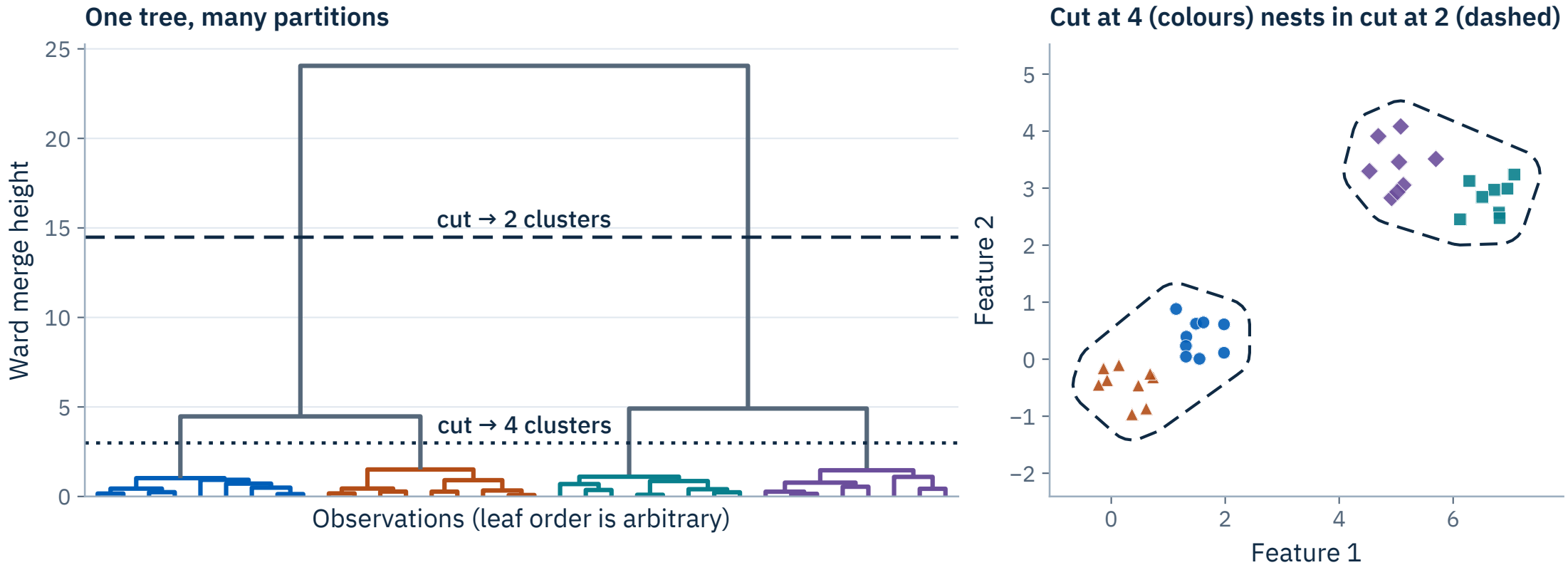
A dendrogram records merge heights



Ward hierarchy of twelve synthetic observations (random_state = 42). Colours mark the three clusters below the dashed cut at height 3.

Cutting the tree at a height gives a partition

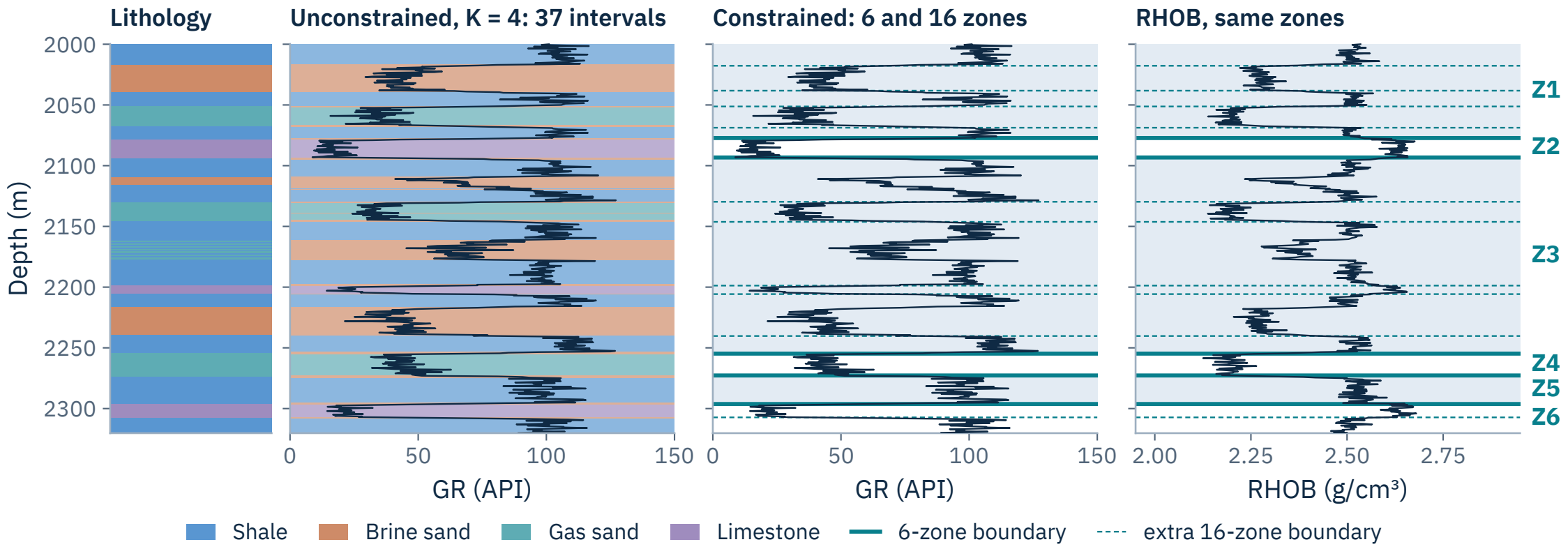
Cuts at different heights give nested partitions of the same tree.



Synthetic 34 points from four blobs (random_state = 1); scipy Ward linkage; cuts halfway between merge heights; fcluster(criterion = "maxclust").

Depth constraints turn clustering into zonation

Merging only adjacent samples gives contiguous zones that nest across scales.



Synthetic well S-1 (seed 7): Ward on four standardized logs, unconstrained (K = 4) and depth-constrained.

Zonation needs only a connectivity matrix

```
from scipy.sparse import diags
from sklearn.cluster import AgglomerativeClustering
from sklearn.preprocessing import StandardScaler

logs = well.sort_values("DEPTH")
Z = StandardScaler().fit_transform(logs[["GR", "RHOB", "NPHI", "LOG_RT"]])
n = len(Z)
A = diags([np.ones(n - 1), np.ones(n - 1)], [-1, 1])  # adjacent samples
zones = AgglomerativeClustering(
    n_clusters=6, linkage="ward", connectivity=A
).fit_predict(Z)
```

Sort by depth first; remove links across logging gaps or well boundaries.

Unconstrained hierarchy grows with n²

Standard hierarchical clustering needs all pairwise distances between samples.

25.0 M

pairwise distances for the 7,071 W5 samples

200 MB

to store them as 8-byte numbers

40 GB

for 100,000 samples

- Subsample, cluster summaries (for example many K-means centroids), or add connectivity before running a full hierarchy.

PRACTICE

4 min · Pairs

Which linkage for which task?

Choose a linkage or constraint for each task and justify it.

1. Separate two dense sand bodies joined by a thin streak of transitional samples.
2. Follow a long, narrow channel feature traced in a seismic attribute map.
3. Build contiguous layers from logs for a reservoir model.

PART 5 OF 6

05

Gaussian mixture models

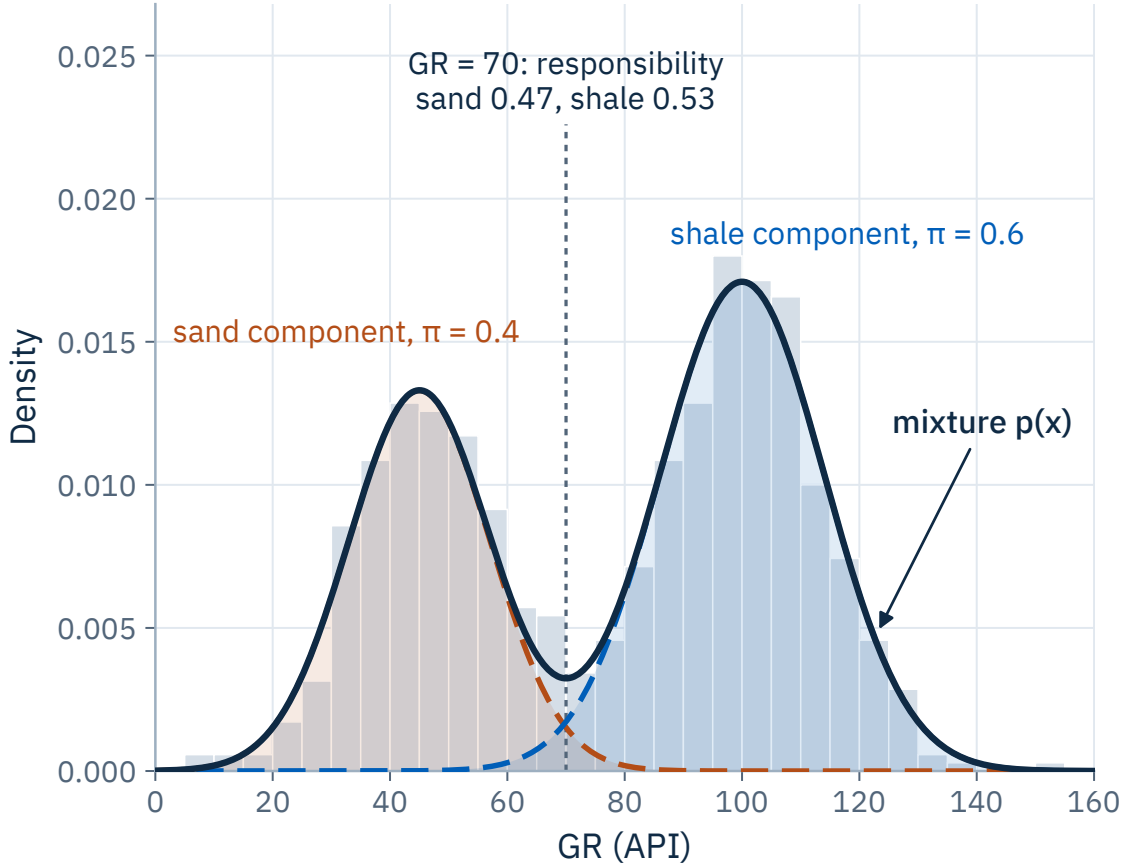
Soft membership allows uncertainty near overlapping groups.

- The mixture density and responsibilities
- Expectation-maximization
- Covariance types and BIC
- Uncertainty maps

The mixture density adds weighted Gaussians

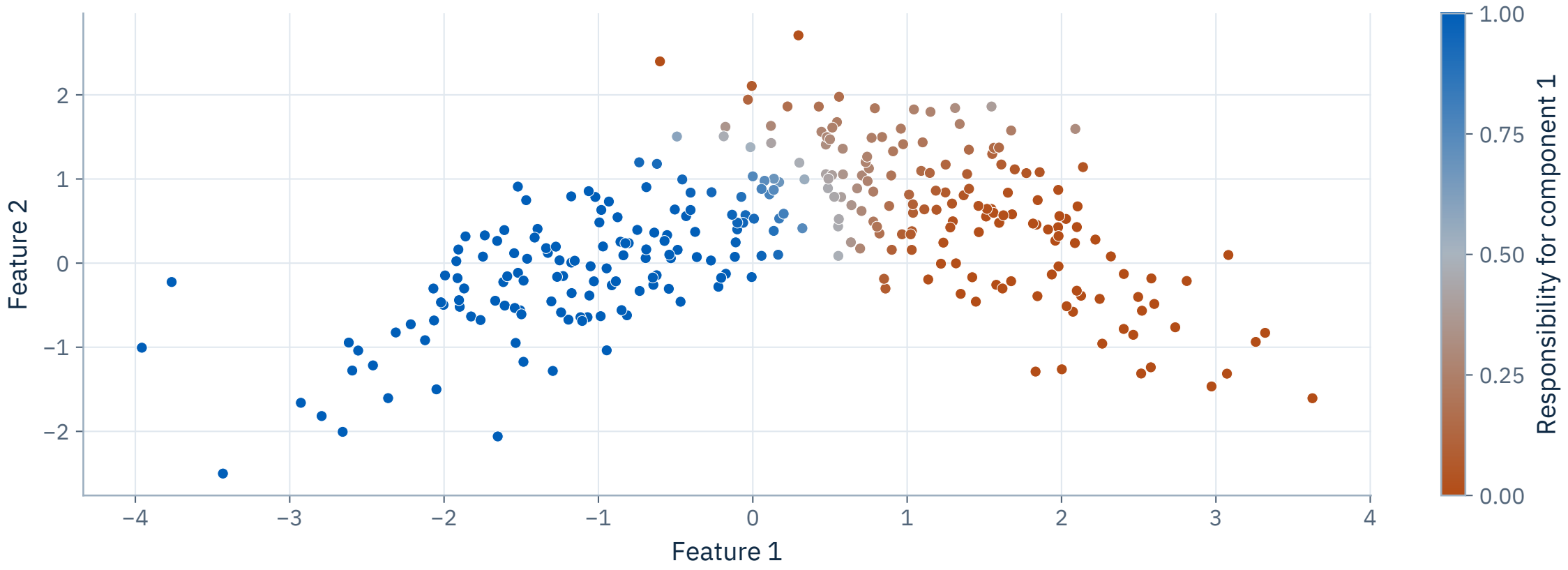
$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

- π_k: mixing weights, non-negative, summing to one
- μ_k and Σ_k: mean and covariance of component k
- **Responsibility** r_{ik}: component k's share of the density at x_i



Exact two-component normal mixture (weights 0.4 and 0.6; means 45 and 100 API; SD 12 and 14 API) with a synthetic histogram (seed 9).

Soft assignments expose overlap



Synthetic overlapping groups (seed 42); two-component Gaussian mixture. Colour: responsibility for component 1; grey points are near 0.5, the ambiguous ones.

Expectation-maximization alternates two steps

Step	Quantity updated	Intuition
E step	Responsibilities	How much does each component explain each observation?
M step	Weights, means, covariances	Refit components using fractional memberships

Repeat until the likelihood improvement is small, with safeguards such as reg_covar.

EM moves the components until the likelihood levels off

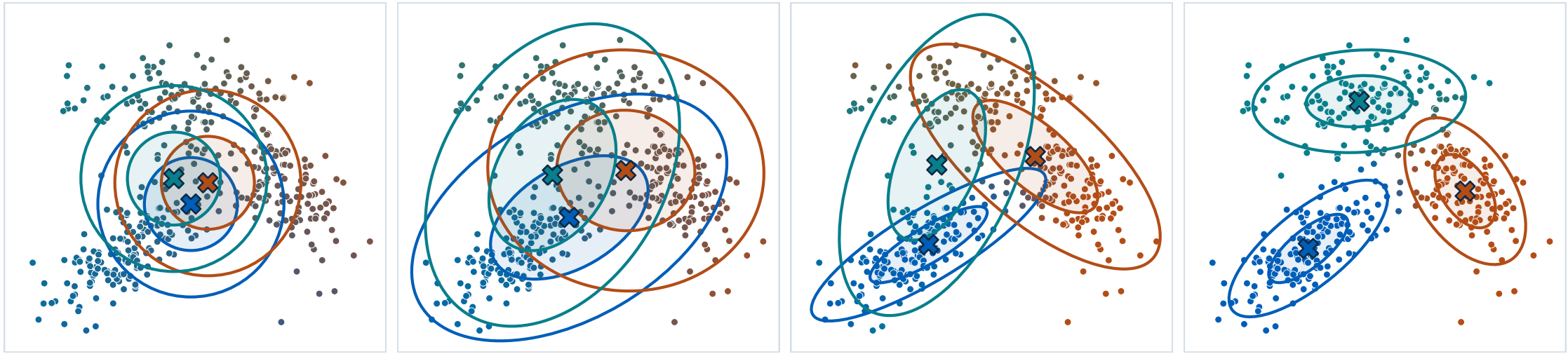
E step: recompute responsibilities. M step: refit weights, means and covariances.

Start
mean log-likelihood −4.433

Iteration 1
mean log-likelihood −3.829

Iteration 5
mean log-likelihood −3.503

Iteration 40
mean log-likelihood −3.311



Synthetic three-component data (seed 31); EM for a full-covariance mixture written in numpy from a deliberately poor start. Point colour blends component colours by responsibility.

PRACTICE

4 min · Individual

A weighted mean update

$$\mu_k^{\text{new}} = \frac{\sum_i r_{ik}\, x_i}{\sum_i r_{ik}}$$

Two observations, x = 0 and x = 10, have responsibilities 0.8 and 0.2 for component 1.

1. Compute the new mean of component 1.
2. Compute the new mean of component 2, whose responsibilities are 0.2 and 0.8.
3. What are the new mixing weights, and the new variance of component 1?

Covariance structure controls flexibility

Type	Assumption	Covariance parameters	Trade-off
Spherical	One variance per component	K	Few parameters, round groups
Diagonal	No within-component correlation	K·d	Axis-aligned ellipses
Tied	One shared full covariance	d(d+1)/2	Same shape and tilt for all
Full	Own full covariance each	K·d(d+1)/2	Flexible, needs more data

Add K·d mean parameters and K – 1 weights to get the total that BIC penalizes.

Covariance type sets the shape of each component

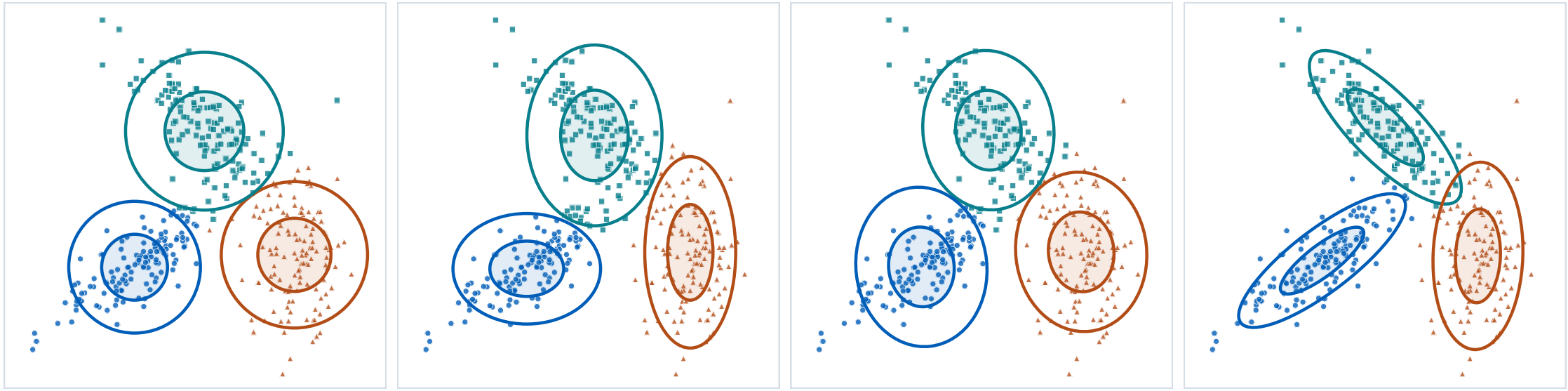
Full covariance costs 3 more parameters than diagonal, yet BIC prefers it clearly.

spherical: 11 parameters
BIC 3086

diag: 14 parameters
BIC 3039

tied: 11 parameters
BIC 3083

full: 17 parameters
BIC 2825

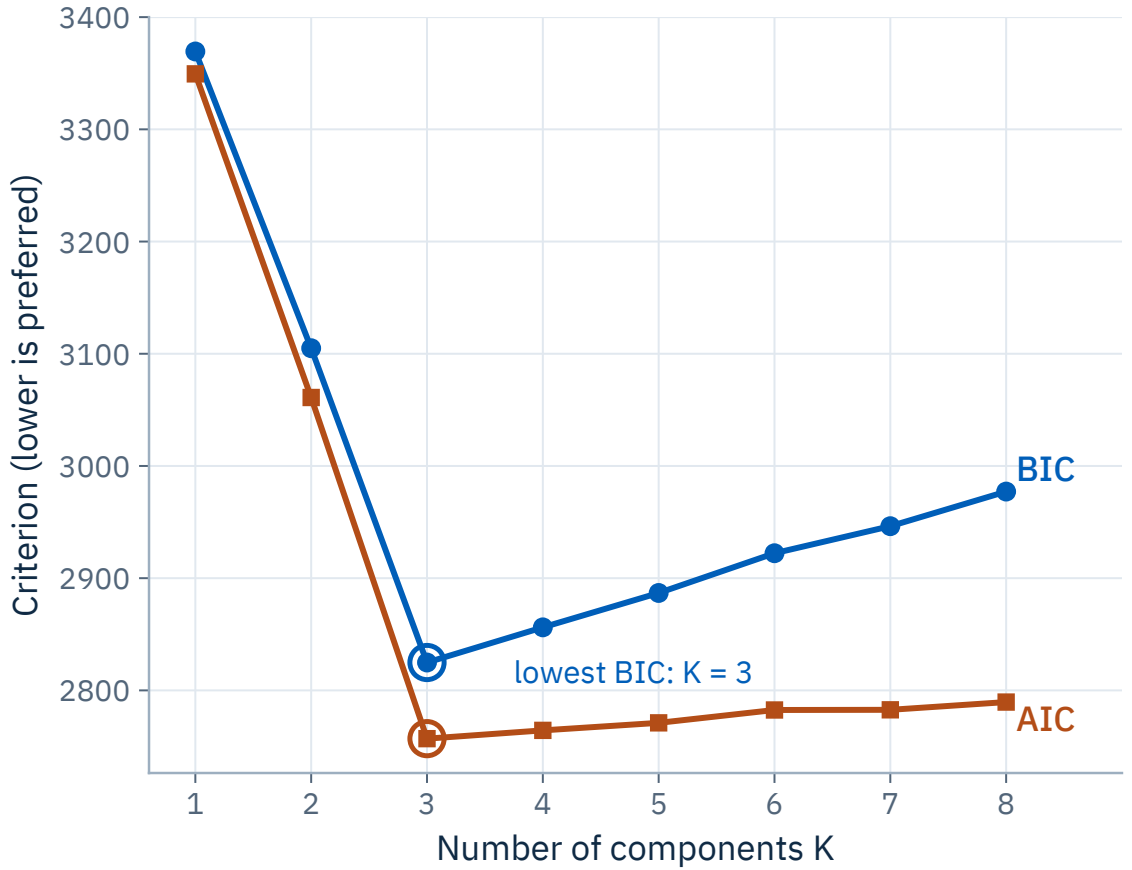


Synthetic three-component data (seed 17); GaussianMixture(K = 3, n_init = 5, random_state = 0) with each covariance_type; ellipses at 1 and 2 standard deviations.

BIC and AIC penalize model complexity

$$\mathrm{BIC} = -2 \log \hat{L} + p \log n$$

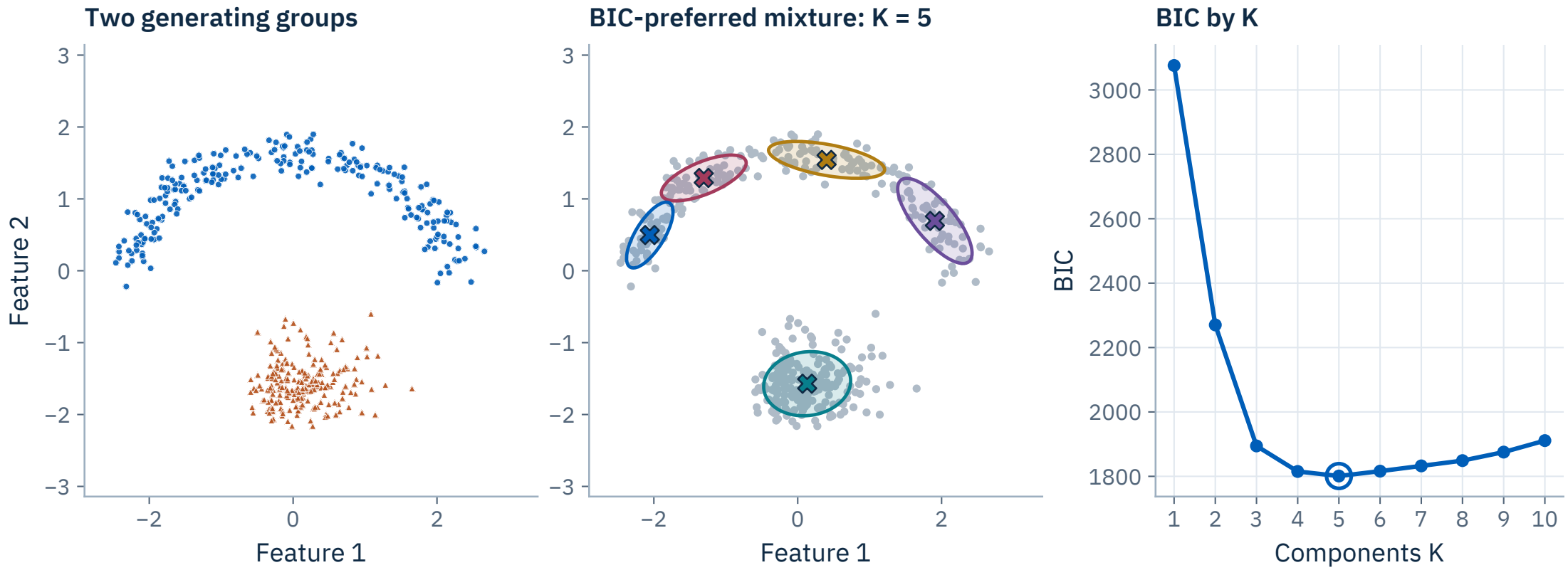
- $\hat{L}$: maximized likelihood; p: free parameters; n: observations
- Lower is preferred; compare models fitted to the same data and features
- AIC replaces $p \log n$ with $2p$, a lighter penalty
- Check stability and plausibility alongside the criterion



Same synthetic data as the covariance figure (three generating components); full-covariance mixtures, K = 1 to 8.

Mixture components are not the same as groups

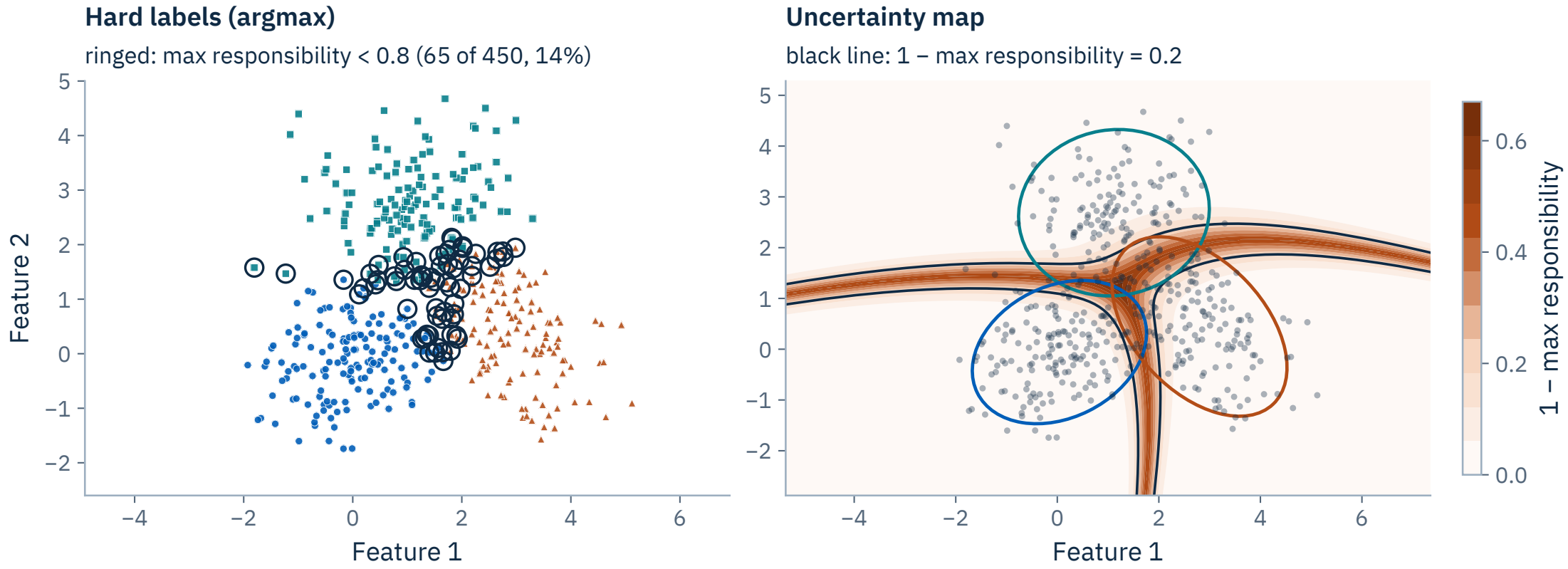
Two generating groups, but BIC prefers five components: Gaussians tile the curved group.



Synthetic: a curved group and a log-normal (skewed) group, seed 23; full-covariance mixtures, n_init = 5; ellipses at 1.5 standard deviations.

Uncertainty maps show where labels are fragile

Responsibilities stay low along boundaries, where a hard label hides ambiguity.



Synthetic overlapping groups (seed 41); GaussianMixture(K = 3, full, n_init = 5, random_state = 0); 2-SD ellipses of the fitted components.

A reproducible mixture fit

```
from sklearn.mixture import GaussianMixture
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler().fit(X_train)
Z_train = scaler.transform(X_train)
gmm = GaussianMixture(
    n_components=3, covariance_type="full",
    n_init=5, reg_covar=1e-6, random_state=42
).fit(Z_train)
membership = gmm.predict_proba(scaler.transform(X_new))
```

PRACTICE

5 min · Pairs

Count the parameters before you fit

Lab 3 fits mixtures to d = 10 standardized oxides on n = 1,684 samples.

1. How many free parameters does a full-covariance mixture with K = 8 have?
2. How many does a diagonal-covariance mixture with K = 8 have?
3. By how much must the full model raise the log-likelihood to beat diagonal on BIC?

PART 6 OF 6

06

Validation and interpretation

A grouping earns trust through several independent kinds of evidence.

- Internal indices and what they reward
- External indices when labels exist
- Stability under resampling
- Profiles, depth tracks and method choice

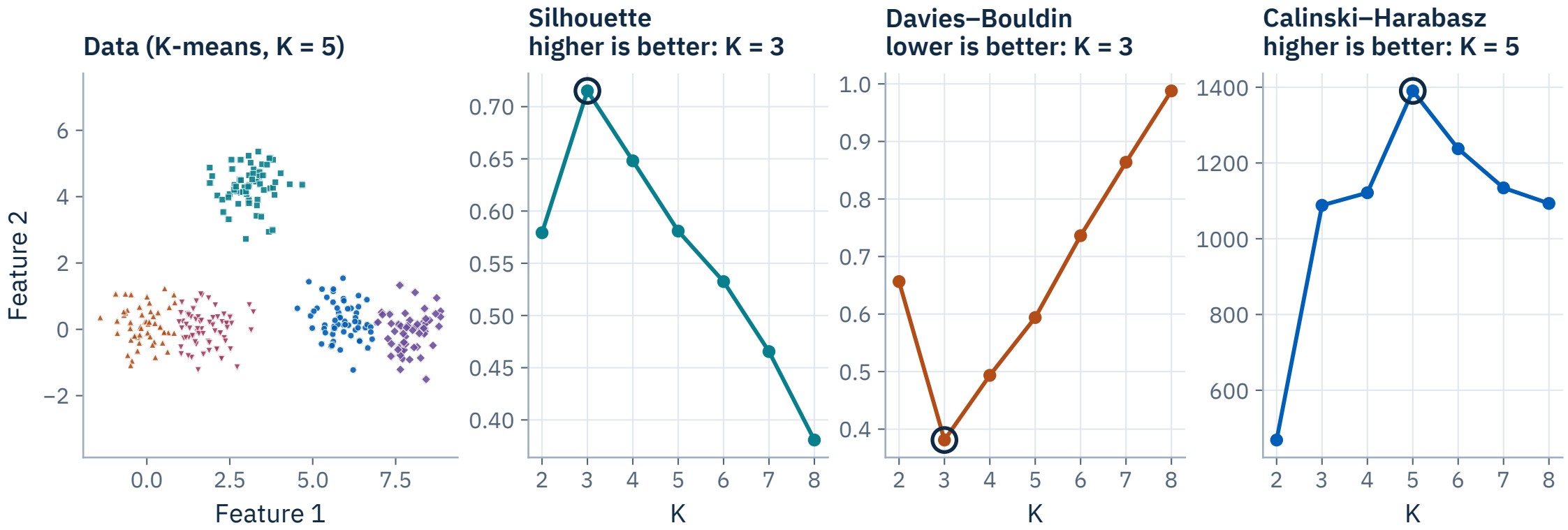
Internal indices reward different geometry

Index	Rewards	Better	Caveat
Silhouette	Points nearer their own cluster than the next	Higher, −1 to 1	Needs all pairwise distances
Davies–Bouldin	Small spread relative to centroid separation	Lower, 0 is best	Centroid-based: favours convex
Calinski–Harabasz	High between- to within-cluster variance	Higher	Favours convex, similar-size

All three use only the data and the labels, so they share the clustering's own assumptions.

Internal indices can disagree on K

Silhouette and Davies–Bouldin see three groups; Calinski–Harabasz resolves all five blobs.



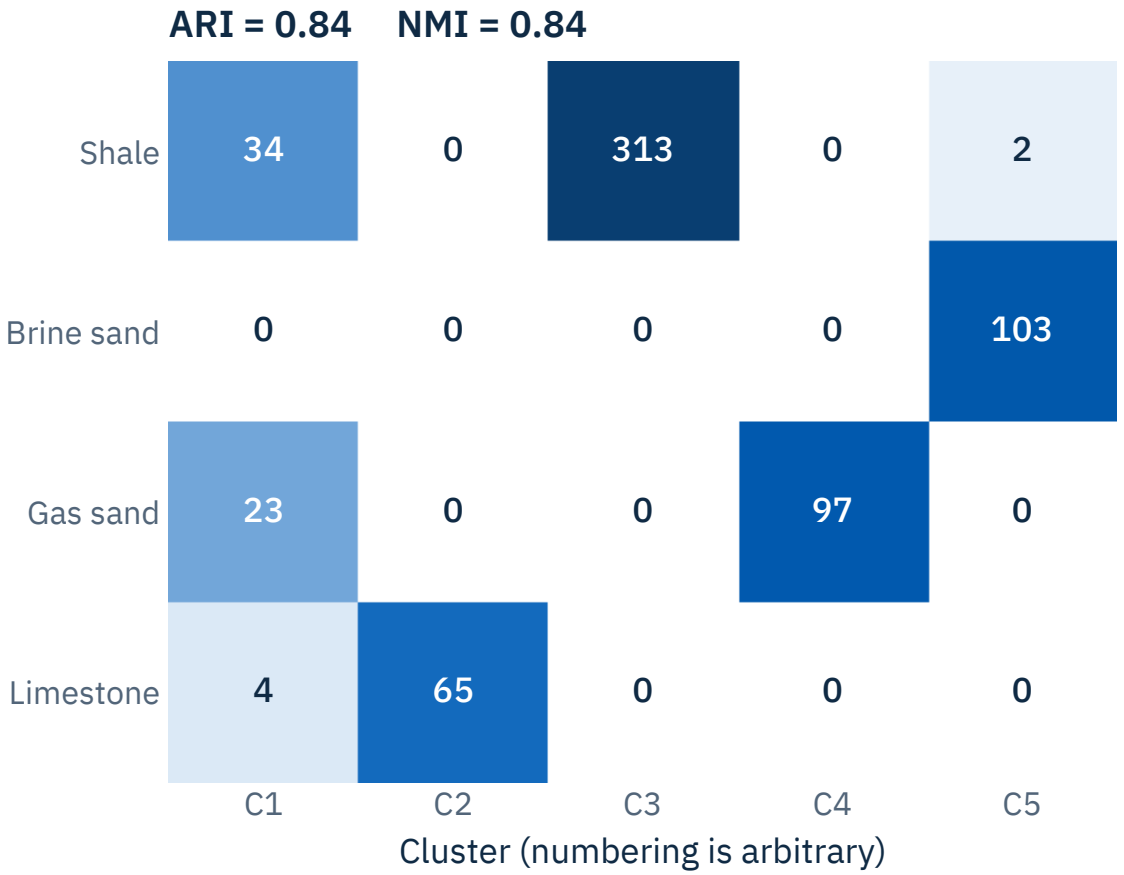
Synthetic five blobs with two close pairs (random_state = 0); K-means (n_init = 10) for K = 2 to 8; scikit-learn metric implementations.

Scores need context

Evidence	What it can establish	What it cannot establish alone
Silhouette	Separation under a selected distance	Geological validity
Adjusted Rand index	Agreement with an external labelling	Correctness of that labelling
Stability across fits	Robustness to perturbations	Usefulness for a decision
Expert review	Domain plausibility	Generalization to all new data

External indices compare clusters with reference labels

- **ARI**: pair agreement corrected for chance; 0 random, 1 identical
- NMI: shared information, 0 to 1; not chance-corrected (AMI is)
- Both ignore cluster numbering; neither says which cluster is wrong
- Use labels to evaluate, never as clustering features



Synthetic well S-1 (seed 7): reference lithology against K-means (K = 5) on standardized logs; counts of depth samples.

Stable clusters survive resampling

- Refit on bootstrap resamples; compare labels on all samples with ARI
- Keep K, features and scaling fixed
- A split that flips between runs is not a finding
- Resample by well or depth block when samples are correlated



Synthetic well S-1 (seed 7), standardized logs; K-means refitted on 30 bootstrap resamples per K; labels predicted for all samples and compared with the full-data fit.

Profile clusters in original units

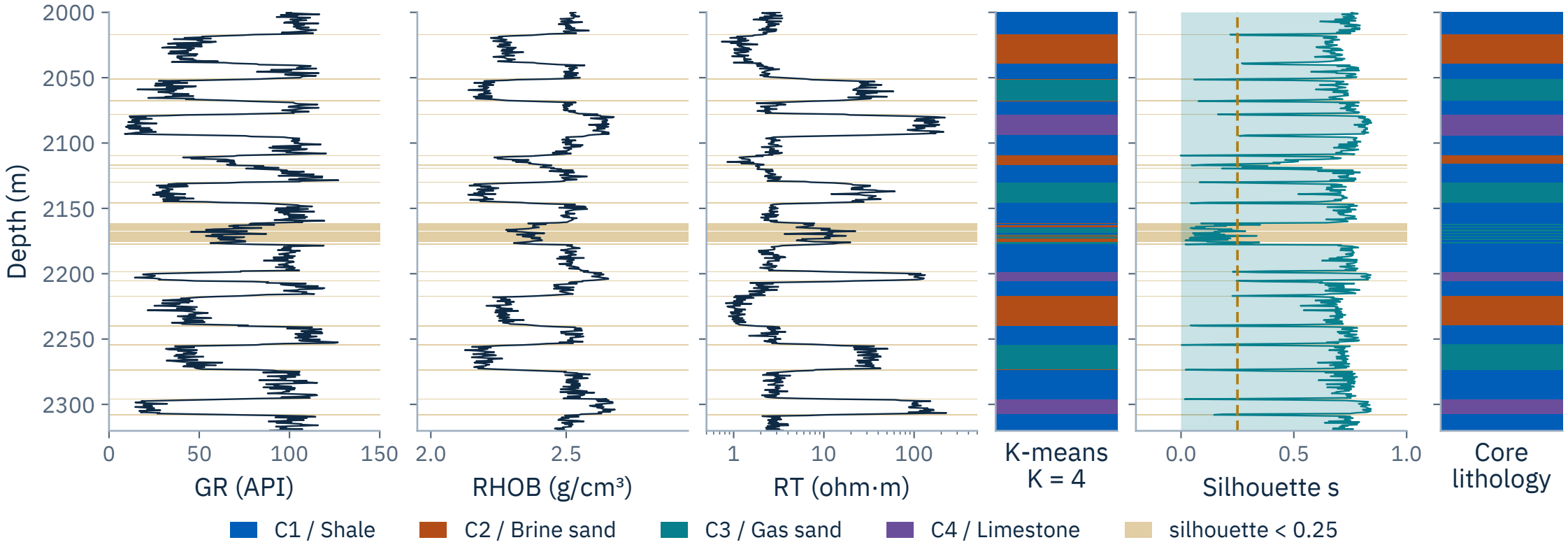
- Report cluster means or medians in physical units
- Add counts, spreads and depth ranges
- Write each reading as a hypothesis to test
- Check against core, cuttings or tests before naming

	GR (API)	RHOB (g/cm³)	NPHI (v/v)	RT (ohm·m)	hypothesis to test
C1 n=332	101	2.51	0.31	2.8	*high GR, low RT: shale-like*
C2 n=125	52	2.30	0.25	1.6	*low GR, low RT: wet-sand-like*
C3 n=115	43	2.22	0.17	25.5	*low GR, high RT, low NPHI: gas-sand-like*
C4 n=69	23	2.63	0.07	103.6	*high RHOB, low NPHI: tight-carbonate-like*

Synthetic well S-1: K-means (K = 4). Cell values are cluster means in original units (RT as geometric mean); colour shows the standardized mean.

Map clusters back onto depth

Low silhouette marks thin beds and boundaries, where one label per sample fails.



Synthetic well S-1 (seed 7): K-means (K = 4) on four standardized logs; per-sample silhouette; core lithology = simulation truth.

Other methods for other shapes and scales

Spectral clustering

- Clusters a similarity graph using its eigenvectors
- Handles non-convex shapes; memory grows with n^2

Mean shift

- Moves every point uphill to a density peak
- Finds K from a bandwidth; slow on large n

BIRCH

- One-pass tree of compact sub-cluster summaries
- Large or streaming data, then a final clustering

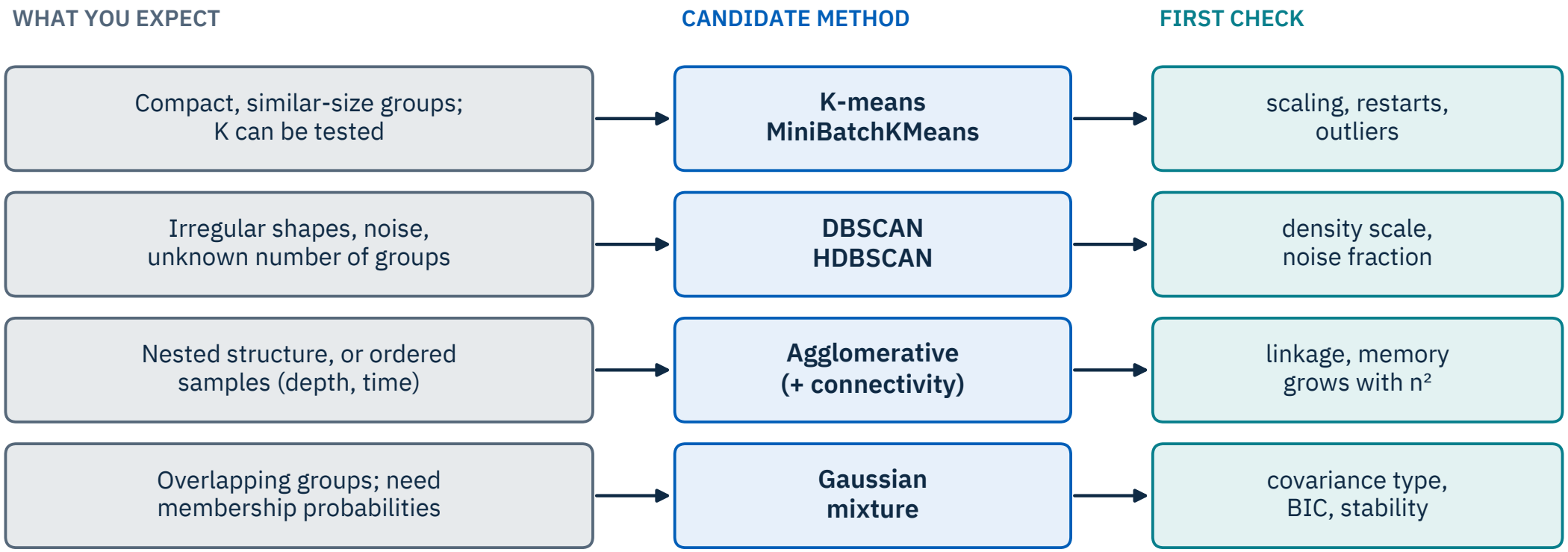
Time series with DTW

- Dynamic time warping aligns shifted profiles
- Pair with k-medoids or a hierarchy (e.g. tslearn)

◆ Choose by the shape, scale and data type you expect, as with every method today.

Select a method by the structure you expect

Start from the structure you expect, then check the method's main assumption first.



Schematic shortlist for a first attempt; not a rule.

PRACTICE

6 min · Groups

Weigh the evidence for a grouping

K-means on five logs from eight wells gives K = 5, mean silhouette 0.46. Bootstrap by well gives median ARI 0.62. Cluster 4 comes almost entirely from one well. ARI with core facies in two cored wells is 0.35.

1. Which evidence supports the grouping, and which undermines it?
2. What is the most likely explanation for cluster 4?
3. What would you do next?

LABORATORY

40 min

Lab 3: Gaussian mixtures on rock geochemistry

- [ ] Read the missing-value report (1,684 of 2,191 rows remain), then standardize.
- [ ] Fit a 3-component mixture; inspect predict and predict_proba.
- [ ] Compare BIC and AIC for K = 1–10 and refit at your chosen K.
- [ ] Crosstab components against ROCK1, compute ARI, and compare covariance types.
- [ ] Extension: the hierarchical (04) or method-comparison (05) notebook.

NOTEBOOK

`exercises/03_gmm/GMM_Rock_Decomposition_Student.ipynb`

EVIDENCE TO SUBMIT

- BIC and AIC curves with your chosen K
- A crosstab with ARI and silhouette
- One paragraph on what the components do and do not represent

DISCUSSION

Lab 3 debrief: BIC fits the density, not the labels

Covariance type, K = 8	BIC	ARI with rock type	Silhouette
Full	18,074	0.28	0.09
Tied	29,432	0.32	0.25
Diagonal	24,776	0.28	0.17
Spherical	31,000	0.30	0.19

The BIC-preferred model describes the oxide distribution best, yet agrees only weakly with the named rock types.

Key takeaways

- ✓ Similarity is a choice: features, transforms and scaling decide the clusters.
- ✓ K-means needs k-means++, several restarts and more than one kind of evidence for K.
- ✓ Density methods find shapes and noise; use a k-distance plot, or HDBSCAN.
- ✓ Hierarchies give nested partitions; connectivity turns them into zonation.
- ✓ Mixtures give soft membership; BIC counts components, not rock types.
- ✓ Validate with stability and external labels; interpret in original units and along depth.

PRACTICE

7 min · Individual

Exit check

1. Why can scaling change the scientific conclusion of a clustering?
2. Why can DBSCAN noise include valid observations?
3. What information disappears when a mixture becomes a hard partition?
4. Name two independent kinds of evidence you need before calling a cluster a facies.

References and further study

- scikit-learn User Guide: clustering, Gaussian mixture models, and clustering performance evaluation.
- K-means: Lloyd (1982), least-squares quantization; Arthur and Vassilvitskii (2007), k-means++.
- Density: Ester et al. (1996), DBSCAN; Ankerst et al. (1999), OPTICS; Campello et al. (2013), HDBSCAN.
- Hierarchy and mixtures: Ward (1963); Dempster, Laird and Rubin (1977), EM; Schwarz (1978), BIC.
- Validation: Rousseeuw (1987), silhouettes; Hubert and Arabie (1985), adjusted Rand index; Hastie, Tibshirani and Friedman, *The Elements of Statistical Learning*, chapter 14.